# Supplemental Material for: Dequantizing Shallow Equivariant Quantum Graph Networks: Periodic Inductive Biases in Molecular Message Passing

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## SM A Experimental Setup, Training Protocol, and Reproducibility

**Training Protocol and Optimization Dynamics.** All baseline and proposed architectures share a standardized training protocol to ensure rigorous iso-capacity evaluation (Table S1). Optimization is conducted using Adam ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$) with an initial learning rate of $1 \times 10^{-3}$, cosine annealing decay down to $\eta_{\min} = 10^{-5}$ across training epochs, $L_2$ weight decay of $10^{-5}$, and global gradient clipping at a maximum $L_2$ norm of 5.0.

For standard benchmark evaluations ($N = 1{,}000$), models are trained for 15 epochs with mini-batch size 16; for full-scale QM9 experiments ($N = 110{,}000$), models are trained for 20 epochs with mini-batch size 64. Training uses a fixed epoch budget without early stopping; final evaluation uses the model checkpoint that achieved the lowest Mean Squared Error (MSE) on the validation partition (250 molecules for standard benchmarks, 10,000 molecules for full QM9). All regression targets $y$ are standardized per target using training set statistics ($y_{\text{norm}} = (y - \mu)/\sigma$) and inverse-transformed for test evaluation, reporting Mean Absolute Error (MAE) in physical units (eV for electronic energies, Debye for dipole moments, and Bohr$^3$ for polarizability).

**Geometric Graph Construction and Radial Encodings.** Molecular graphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ are constructed by placing edges between atom pairs with Euclidean distance $r_{ij} \leq 5.0$ Å, supplemented with a $k$-nearest neighbor fallback ($k = 8$) to guarantee connectivity for isolated peripheral atoms. Interatomic distances are encoded using $K = 8$ Gaussian Radial Basis Functions (RBF):

$$\phi_k(d_{ij}) = \exp\left(-\gamma_k(d_{ij} - \mu_k)^2\right), \quad k \in \{1, \ldots, K\} \quad (1)$$

where $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|^2$ is the squared Euclidean distance, with centers $\mu_k$ evenly spaced across $[0, 10]$ Å$^2$. Because $\sqrt{10} \approx 3.16$ Å, the RBF centers intentionally concentrate radial resolution within the primary covalent and hydrogen bonding shells ($\leq 3.16$ Å), where orbital overlaps are most sensitive, while the broader 5.0 Å graph radius cutoff preserves the long-range topology of the molecular scaffold.

**Computational Hardware and Profiling Environments.** Standard model training and benchmark sweeps were executed on NVIDIA A100 (40GB) GPUs. To isolate algorithmic complexity from hardware acceleration and eliminate GPU kernel dispatch variance during wall-time scaling profiling (SM F), Q-PRISM PennyLane statevector simulations and PRISM classical forward/backward passes were profiled on an isolated single CPU core (AMD EPYC 7742 @ 2.25 GHz).

**Code and Data Availability.** All source code for Q-PRISM, PRISM, and the complete experimental evaluation pipeline (training loops, analytical noise modeling, statistical significance testing, and visualization scripts) will be made publicly available in an open-source repository upon publication. The QM9 benchmark dataset [18] is publicly accessible via PyTorch Geometric. The MD17 conformational dataset [7] and 5-particle Coulomb N-body trajectory simulations are generated using established protocols documented in the literature.

## SM B Statistical Methodology, Power Analysis, and Multiplicity Control

**Random Seed Protocol and Variance Decomposition.** All experimental random seeds (42–46 for 5-seed benchmarks; 42–61 for 20-seed ablations) independently govern both the initialization of trainable network weights and the pseudo-random partition of dataset molecules into training, validation, and test splits. Consequently, the reported standard deviations capture composite empirical variance across optimization landscapes

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Table S1: **Training Hyperparameters** for all models across QM9 benchmark experiments. All models share identical training protocol unless otherwise noted. Hyperparameters were fixed *a priori* and not tuned on the test set.

| Hyperparameter | Value | Notes |
|---|---|---|
| *Optimization* | | |
| Optimizer | Adam | $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ |
| Learning Rate | $1 \times 10^{-3}$ | Fixed for all models |
| LR Scheduler | CosineAnnealingLR | $T_{\max} = 15$ epochs, $\eta_{\min} = 10^{-5}$ |
| Weight Decay | $10^{-5}$ | $L_2$ regularization |
| Gradient Clipping | Max norm = 5.0 | Applied globally across all parameters |
| *Training Protocol* | | |
| Epochs | 15 | Standard benchmark ($N = 1{,}000$); 20 epochs for full QM9 |
| Batch Size | 16 | Standard benchmark ($N = 1{,}000$); 64 for full QM9 |
| Early Stopping | None | Fixed epoch budget; end-of-training checkpoint |
| Validation Set | 250 molecules | Evaluated during training |
| Loss Function | Mean Squared Error (MSE) | Computed on standardized targets ($y_{\text{norm}} = (y - \mu)/\sigma$) |
| *Data Preprocessing* | | |
| Target Normalization | Per-target mean/std | Computed on training set; inverse-transformed for evaluation (MAE in eV) |
| Radius Cutoff | 5.0 Å | For edge construction ($k$-NN fallback: $k = 8$) |
| RBF Centers | $K = 8$ | Evenly spaced in $[0, 10]$ Å$^2$ (centers $\leq 3.16$ Å target primary bonding shell; 5.0 Å cutoff retains graph topology) |
| *Architecture (shared across all iso-parameter models)* | | |
| GNN Layers ($T$) | 2 | Message-passing iterations |
| Node Embedding Dim ($d_{\text{node}}$) | 11 | One-hot atomic number (H, C, N, O, F: 5) + degree features (6) |
| Edge Head Input Dim ($d_{\text{edge}}$) | 13 | Projected node features (4) + RBF (8) + bond angle (1) |
| Hidden Dim ($d_h$) | 4 | Edge head hidden dim; scaled from 4 (XS) to 128 (XXL) in Table S8 |
| Readout | Global mean pooling | $\bar{\mathbf{h}} = \frac{1}{\lvert V \rvert} \sum_i \mathbf{h}_i^{(T)}$ |
| Output Head | Linear($d_{\text{hidden}}$, 1) | Single scalar prediction |
| *PRISM-Specific* | | |
| SIREN $\omega_0$ | 30.0 | Fixed *a priori* per Sitzmann et al. [25] |
| SIREN Init ($\mathbf{W}_1$) | $\mathcal{U}(-1/d_{\text{edge}}, 1/d_{\text{edge}})$ | First layer ($d_{\text{edge}} = 13$; Sitzmann et al. [25]) |
| SIREN Init ($\mathbf{W}_2$) | $\mathcal{U}(-\sqrt{6/d_h}/\omega_0, \sqrt{6/d_h}/\omega_0)$ | Hidden layer ($d_h = 4$; Sitzmann et al. [25]) |
| *Q-PRISM-Specific (PennyLane Simulation)* | | |
| Simulator Backend | `default.qubit` | Statevector (exact expectation values, infinite shots) |
| Qubit Count | 4 | Per PQC head |
| PQC Layers ($L$) | 2 | Default; varied in circuit scaling study |
| Differentiation | Backpropagation | Via PennyLane autograd interface |
| *Computational Resources* | | |
| Hardware | NVIDIA A100 (40GB) / CPU | GPU for model training; single CPU core for Q-PRISM vs. PRISM wall-time scaling profiling |
| Training Time (PRISM) | $\sim 20$ s / seed | $N = 1{,}000$, 15 epochs |
| Training Time (Q-PRISM) | $\sim 3$ min / seed | $N = 1{,}000$, 15 epochs (PennyLane statevector simulation) |
| Random Seeds | 42–46 (5-seed); 42–61 (20-seed) | Control both init and data split |

and molecular data sampling. The stability of effect sizes across 20 independent splits (Cohen's $d > 0.85$ on HOMO) establishes that performance differences reflect intrinsic architectural properties rather than sample partitioning artifacts.

**Statistical Power and Non-Parametric Testing Framework.** In the 5-seed benchmarks, non-parametric Wilcoxon signed-rank tests serve as indicators of directional sign consistency across splits (where unanimous agreement yields the minimum achievable two-sided value $p = 0.0625$ at $n = 5$). Formal non-parametric hypothesis testing is powered by the 20-seed ablation (main text Section 4.3), which provides sufficient statistical power to resolve rank-based significance at $\alpha = 0.05$. Post-hoc power analyses confirm that the 5-seed protocol ($n = 5$, paired) achieves 80% statistical power ($1 - \beta = 0.80$) for large effect sizes of Cohen's $d \geq 1.53$ at $\alpha = 0.05$, whereas the 20-seed protocol achieves 80% power for medium-large effects ($d \geq 0.66$), matching the observed effect size ($d = 0.85$) for the two-qubit entanglement ablation on HOMO.

**Multiplicity Control Framework.** A critical methodological requirement in multi-target molecular benchmarking is controlling for false discoveries arising from multiple hypothesis testing. Evaluating 8 model architectures across 6 physical targets yields $m = 8 \times 6 = 48$ pairwise statistical comparisons, wherein each architecture is compared against its designated classical baseline (Classical Twin tanh-MLP) via paired two-tailed $t$-tests across 5 random seeds. The 48 hypotheses encompass all pairwise model-vs-baseline comparisons across the full QM9 target spectrum (HOMO, LUMO, Gap, Dipole Moment, Polarizability, and Internal Energy $U_0$).

To prevent false discovery inflation while maintaining statistical power, we report two multiplicity control standards:

1. **Family-Wise Error Rate (FWER) via Bonferroni Correction:**

$$p_{\text{Bonf}} = \min(1.0, m \cdot p_{\text{raw}}) \tag{2}$$

guaranteeing strict family-wise error rate control at level $\alpha = 0.05$.

2. **False Discovery Rate (FDR) via Benjamini–Hochberg (BH):**

$$q_{\text{FDR}} = \min_{k \geq i} \left( \frac{m}{k} p_{(k)} \right) \tag{3}$$

Table S2: **Multiplicity-Corrected Statistical Significance for PRISM and Q-PRISM** across QM9 targets ($m = 48$ hypotheses). Mean difference ($\Delta = \text{MAE}_{\text{Model}} - \text{MAE}_{\text{Twin}}$), raw paired $t$-test $p$, Benjamini–Hochberg FDR $q$-value, and Bonferroni $p_{\text{Bonf}}$.

| Comparison | Target | $\Delta$ | $p_{\text{raw}}$ | $q_{\text{FDR}}$ | $p_{\text{Bonf}}$ |
|---|---|---|---|---|---|
| *PRISM (SIREN) vs. Classical Twin* (tanh-*MLP*) | | | | | |
| | HOMO (eV) | $-0.0746$ | $9.57 \times 10^{-3}$ | $\mathbf{0.0184}^{\dagger}$ | 0.459 |
| | LUMO (eV) | $-0.2405$ | $3.67 \times 10^{-4}$ | $\mathbf{0.0016}^{\dagger\dagger}$ | $\mathbf{0.0176}^{*}$ |
| | Gap (eV) | $-0.2489$ | $3.47 \times 10^{-4}$ | $\mathbf{0.0016}^{\dagger\dagger}$ | $\mathbf{0.0166}^{*}$ |
| | Dipole (D) | $-0.1927$ | $8.31 \times 10^{-3}$ | $\mathbf{0.0173}^{\dagger}$ | 0.399 |
| | Polariz. ($\alpha$) | $-0.4253$ | $2.22 \times 10^{-2}$ | $\mathbf{0.0355}^{\dagger}$ | 1.000 |
| *Q-PRISM (PQC) vs. PRISM (SIREN)* | | | | | |
| | HOMO (eV) | $+0.0334$ | $4.32 \times 10^{-4}$ | $\mathbf{0.0017}^{\dagger\dagger}$ | $\mathbf{0.0207}^{*}$ |
| | LUMO (eV) | $+0.1484$ | $1.93 \times 10^{-3}$ | $\mathbf{0.0052}^{\dagger\dagger}$ | 0.0926 |
| | Gap (eV) | $+0.1634$ | $5.79 \times 10^{-3}$ | $\mathbf{0.0126}^{\dagger}$ | 0.2777 |
| | Dipole (D) | $+0.0526$ | $3.11 \times 10^{-3}$ | $\mathbf{0.0075}^{\dagger\dagger}$ | 0.1494 |
| *Q-PRISM (PQC) vs. Classical Twin* (tanh-*MLP*) | | | | | |
| | LUMO (eV) | $-0.0921$ | $9.21 \times 10^{-3}$ | $\mathbf{0.0184}^{\dagger}$ | 0.4422 |
| | Gap (eV) | $-0.0855$ | $1.64 \times 10^{-2}$ | $\mathbf{0.0272}^{\dagger}$ | 0.7882 |

controlling the expected proportion of false positive discoveries at level $\alpha = 0.05$.

**Analysis of Multiplicity-Corrected Findings.** As summarized in Table S2 and detailed across all 48 hypotheses in Table S3:

- PRISM's performance advantage over the Classical Twin (tanh) remains statistically significant after Benjamini–Hochberg FDR correction across 5 out of 6 QM9 targets ($q_{\text{FDR}} < 0.02$ for HOMO, LUMO, Gap, Dipole; $q_{\text{FDR}} = 0.0355$ for $\alpha$). On LUMO and Gap, the advantage survives even conservative Bonferroni correction ($p_{\text{Bonf}} = 0.0176$ and 0.0166).
- PRISM is statistically significantly superior to Q-PRISM across all four intensive electronic and dipole targets ($q_{\text{FDR}} \leq 0.0126$), demonstrating that the classical trigonometric representation provides a more effective functional inductive bias than the 4-qubit PQC.
- At $n = 5$, the minimum attainable two-sided $p$-value for the non-parametric Wilcoxon signed-rank test is $p = 0.0625$ (corresponding to unanimous rank agreement across all 5 seeds). We report these values in Table S3 to confirm directional consistency across all splits. Formal non-parametric hypothesis testing with full statistical power is conducted via the 20-seed ablation (main text Section 4.3), which achieves statistically significant separation ($p < 0.05$) under both parametric paired $t$-tests and non-parametric Wilcoxon signed-rank tests.

**Complete 48-Hypothesis Multiplicity Report.** To provide full transparency on multiple hypothesis testing across the target hierarchy, Table S3 details the complete statistical evaluation across all $m = 48$ pairwise comparisons (8 architectures × 6 QM9 targets, evaluated across 5 independent random splits and initializations).

# SM C Mathematical Proofs and Algorithmic Specifications

**Theorem 1** (E(3) Equivariance of Q-PRISM Layer)**.** *Let $\mathcal{T}_{\mathbf{Q},\mathbf{g}}(\mathbf{x}) = \mathbf{x}\mathbf{Q}^T + \mathbf{g}$ denote an arbitrary continuous Euclidean transformation with rotation/reflection $\mathbf{Q} \in \mathrm{O}(3)$ and translation $\mathbf{g} \in \mathbb{R}^3$.*[1] *Then for the Q-PRISM layer defined in main text Eqs. (9–11):*

$$\mathbf{h}_i^{l+1}(\mathcal{T}_{\mathbf{Q},\mathbf{g}}(\mathbf{x}), \mathbf{h}) = \mathbf{h}_i^{l+1}(\mathbf{x}, \mathbf{h}) \quad (\textit{Invariance}) \tag{4}$$

$$\mathbf{x}_i^{l+1}(\mathcal{T}_{\mathbf{Q},\mathbf{g}}(\mathbf{x}), \mathbf{h}) = \mathcal{T}_{\mathbf{Q},\mathbf{g}}(\mathbf{x}_i^{l+1}(\mathbf{x}, \mathbf{h})) \quad (\textit{Equivariance}) \tag{5}$$

*Proof.* Under transformation $\mathcal{T}_{\mathbf{Q},\mathbf{g}}$, the relative coordinate vector transforms as $\mathbf{r}'_{ij} = (\mathbf{x}_j\mathbf{Q}^T + \mathbf{g}) - (\mathbf{x}_i\mathbf{Q}^T + \mathbf{g}) = \mathbf{r}_{ij}\mathbf{Q}^T$, yielding invariant squared Euclidean distance:

$$d'_{ij} = \|\mathbf{r}'_{ij}\|^2 = (\mathbf{r}_{ij}\mathbf{Q}^T)(\mathbf{r}_{ij}\mathbf{Q}^T)^T = \mathbf{r}_{ij}\mathbf{Q}^T\mathbf{Q}\mathbf{r}_{ij}^T = \|\mathbf{r}_{ij}\|^2 = d_{ij} \tag{6}$$

by the orthogonality property $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$.

The Gaussian RBF features $\boldsymbol{\phi}(d_{ij})$ are scalar functions of the invariant squared distance $d_{ij}$ and are thus invariant. The 3-body mean bond angles $\bar{\theta}_{ij}$ depend exclusively on inner products of unit relative vectors (main text Eq. 5), which satisfy:

$$(\hat{\mathbf{r}}_{ij}\mathbf{Q}^T) \cdot (\hat{\mathbf{r}}_{ik}\mathbf{Q}^T) = \hat{\mathbf{r}}_{ij}\mathbf{Q}^T\mathbf{Q}\hat{\mathbf{r}}_{ik}^T = \hat{\mathbf{r}}_{ij} \cdot \hat{\mathbf{r}}_{ik} \tag{7}$$

by orthogonality of $\mathbf{Q}$. Because all inputs $\mathbf{u}_{ij}^{(s)} = [\mathbf{h}_i^p, \mathbf{h}_j^p, \boldsymbol{\phi}(d_{ij}), \bar{\theta}_{ij}]$ to the edge heads are strictly O(3)-invariant, the parameterized quantum state $|\psi_{ij}^{(s)}\rangle$ and all resulting measurement observables ($\mathbf{m}_{ij}$, $\phi_x$) are invariant. The coordinate gating function $\phi_x(\mathbf{m}_{ij}) = \tanh(\mathbf{W}_g[\phi_x^{(1)}, \phi_x^{(2)}]^T)$ is a scalar function of invariant quantities and is thus invariant.

Substituting into the coordinate update equation:

$$\begin{aligned}
\mathbf{x}_i^{l+1}(\mathbf{x}', \mathbf{h}) &= \mathbf{x}'_i + \frac{1}{|\mathcal{N}(i)|}\sum_{j\in\mathcal{N}(i)} \mathbf{r}'_{ij} \cdot \phi_x(\mathbf{m}_{ij}) \\
&= (\mathbf{x}_i\mathbf{Q}^T + \mathbf{g}) + \left[\frac{1}{|\mathcal{N}(i)|}\sum_{j\in\mathcal{N}(i)} \mathbf{r}_{ij}\phi_x(\mathbf{m}_{ij})\right]\mathbf{Q}^T \\
&= \left(\mathbf{x}_i + \frac{1}{|\mathcal{N}(i)|}\sum_{j\in\mathcal{N}(i)} \mathbf{r}_{ij}\phi_x(\mathbf{m}_{ij})\right)\mathbf{Q}^T + \mathbf{g} \\
&= \mathcal{T}_{\mathbf{Q},\mathbf{g}}(\mathbf{x}_i^{l+1}(\mathbf{x}, \mathbf{h}))
\end{aligned} \tag{8}$$

Node feature updates $\mathbf{h}_i^{l+1}$ depend solely on invariant messages $\mathbf{m}_{ij}$ and invariant node states $\mathbf{h}_i^l$, establishing exact feature invariance.[2] □

[1] We adopt a row-vector convention throughout: coordinates are stored as $\mathbf{x}_i \in \mathbb{R}^{1\times 3}$ (row vectors), so the transformation is $\mathbf{x}\mathbf{Q}^T + \mathbf{g}$ rather than $\mathbf{Q}\mathbf{x} + \mathbf{g}$. This matches PyTorch's default batch-of-row-vectors layout.

[2] Our E(3) equivariance formulation treats all spatial vectors as

Table S3: **Complete 48-Hypothesis Multiplicity-Correction and Non-Parametric Validation Report across QM9 Targets.** Paired comparisons across 5 random seeds ($m = 48$ hypotheses). $\Delta = \mathrm{MAE}_A - \mathrm{MAE}_B$. Unadjusted paired Student's $t$-test ($p_t$), two-sided Wilcoxon signed-rank test ($p_{\mathrm{Wilcox}}$), Benjamini-Hochberg False Discovery Rate $q$-value ($q_{\mathrm{FDR}}$), and Bonferroni-adjusted $p$-value ($p_{\mathrm{Bonf}}$). Bold $q_{\mathrm{FDR}}$ indicates significance at $\alpha = 0.05$ ($^{\dagger}q_{\mathrm{FDR}} < 0.05$). Note: with $n = 5$ seeds, the minimum attainable two-sided Wilcoxon $p$-value is $2/32 = 0.0625$ (attained whenever all 5 seeds unanimously favour model $A$ over $B$).

| **Target Property** | **Hypothesis Comparison ($A$ vs. $B$)** | **$\Delta$** | **$p_t$** | **$p_{\mathrm{Wilcox}}$** | **$q_{\mathrm{FDR}}$** | **$p_{\mathrm{Bonf}}$** |
|---|---|---|---|---|---|---|
| HOMO ($\epsilon_{\mathrm{H}}$, eV) | Q-PRISM (PQC) vs. Classical Twin (tanh) | -0.0411 | 0.0573 | 0.0625 | 0.0723 | 1.00 |
| | Q-PRISM (PQC) vs. SchNet | -0.0786 | $8.30 \times 10^{-4}$ | 0.0625 | $\mathbf{3.07 \times 10^{-3}}^{\dagger}$ | 0.0399 |
| | Q-PRISM (PQC) vs. PaiNN | +0.0194 | 0.0921 | 0.1250 | 0.1133 | 1.00 |
| | Q-PRISM (PQC) vs. EGNN-PM | -0.0625 | $1.62 \times 10^{-3}$ | 0.0625 | $\mathbf{4.86 \times 10^{-3}}^{\dagger}$ | 0.0778 |
| | Q-PRISM (PQC) vs. EGNN Std | -0.0531 | 0.0126 | 0.0625 | $\mathbf{0.0216}^{\dagger}$ | 0.6050 |
| | Q-PRISM (PQC) vs. PRISM (SIREN) | +0.0334 | $4.32 \times 10^{-4}$ | 0.0625 | $\mathbf{1.73 \times 10^{-3}}^{\dagger}$ | 0.0207 |
| | Q-PRISM (PQC) vs. Fourier Twin (RFF) | -0.1210 | $1.02 \times 10^{-5}$ | 0.0625 | $\mathbf{1.62 \times 10^{-4}}^{\dagger}$ | $4.87 \times 10^{-4}$ |
| | PRISM (SIREN) vs. Classical Twin (tanh) | -0.0746 | $9.57 \times 10^{-3}$ | 0.0625 | $\mathbf{0.0184}^{\dagger}$ | 0.4594 |
| LUMO ($\epsilon_{\mathrm{L}}$, eV) | Q-PRISM (PQC) vs. Classical Twin (tanh) | -0.0921 | $9.21 \times 10^{-3}$ | 0.0625 | $\mathbf{0.0184}^{\dagger}$ | 0.4422 |
| | Q-PRISM (PQC) vs. SchNet | -0.0647 | $1.21 \times 10^{-3}$ | 0.0625 | $\mathbf{4.16 \times 10^{-3}}^{\dagger}$ | 0.0583 |
| | Q-PRISM (PQC) vs. PaiNN | +0.2298 | $5.12 \times 10^{-5}$ | 0.0625 | $\mathbf{3.51 \times 10^{-4}}^{\dagger}$ | $2.46 \times 10^{-3}$ |
| | Q-PRISM (PQC) vs. EGNN-PM | -0.0677 | $4.14 \times 10^{-3}$ | 0.0625 | $\mathbf{9.47 \times 10^{-3}}^{\dagger}$ | 0.1989 |
| | Q-PRISM (PQC) vs. EGNN Std | -0.0653 | 0.0115 | 0.0625 | $\mathbf{0.0212}^{\dagger}$ | 0.5502 |
| | Q-PRISM (PQC) vs. PRISM (SIREN) | +0.1484 | $1.93 \times 10^{-3}$ | 0.0625 | $\mathbf{5.23 \times 10^{-3}}^{\dagger}$ | 0.0926 |
| | Q-PRISM (PQC) vs. Fourier Twin (RFF) | -0.1477 | $2.79 \times 10^{-4}$ | 0.0625 | $\mathbf{1.49 \times 10^{-3}}^{\dagger}$ | 0.0134 |
| | PRISM (SIREN) vs. Classical Twin (tanh) | -0.2405 | $3.67 \times 10^{-4}$ | 0.0625 | $\mathbf{1.60 \times 10^{-3}}^{\dagger}$ | 0.0176 |
| Gap ($\Delta\epsilon$, eV) | Q-PRISM (PQC) vs. Classical Twin (tanh) | -0.0855 | 0.0164 | 0.0625 | $\mathbf{0.0272}^{\dagger}$ | 0.7882 |
| | Q-PRISM (PQC) vs. SchNet | -0.0479 | 0.1470 | 0.0625 | 0.1641 | 1.00 |
| | Q-PRISM (PQC) vs. PaiNN | +0.2504 | $1.47 \times 10^{-3}$ | 0.0625 | $\mathbf{4.72 \times 10^{-3}}^{\dagger}$ | 0.0708 |
| | Q-PRISM (PQC) vs. EGNN-PM | -0.0482 | 0.1424 | 0.1875 | 0.1628 | 1.00 |
| | Q-PRISM (PQC) vs. EGNN Std | -0.0553 | 0.0489 | 0.0625 | 0.0652 | 1.00 |
| | Q-PRISM (PQC) vs. PRISM (SIREN) | +0.1634 | $5.79 \times 10^{-3}$ | 0.0625 | $\mathbf{0.0126}^{\dagger}$ | 0.2777 |
| | Q-PRISM (PQC) vs. Fourier Twin (RFF) | -0.1256 | $3.12 \times 10^{-3}$ | 0.0625 | $\mathbf{7.49 \times 10^{-3}}^{\dagger}$ | 0.1498 |
| | PRISM (SIREN) vs. Classical Twin (tanh) | -0.2489 | $3.46 \times 10^{-4}$ | 0.0625 | $\mathbf{1.60 \times 10^{-3}}^{\dagger}$ | 0.0166 |
| Dipole Moment ($\mu$, Debye) | Q-PRISM (PQC) vs. Classical Twin (tanh) | -0.1401 | 0.0340 | 0.1250 | $\mathbf{0.0494}^{\dagger}$ | 1.00 |
| | Q-PRISM (PQC) vs. SchNet | -0.1597 | $2.49 \times 10^{-4}$ | 0.0625 | $\mathbf{1.49 \times 10^{-3}}^{\dagger}$ | 0.0120 |
| | Q-PRISM (PQC) vs. PaiNN | -0.0552 | 0.4359 | 0.6250 | 0.4755 | 1.00 |
| | Q-PRISM (PQC) vs. EGNN-PM | -0.1148 | 0.0252 | 0.0625 | $\mathbf{0.0390}^{\dagger}$ | 1.00 |
| | Q-PRISM (PQC) vs. EGNN Std | -0.0868 | 0.0318 | 0.0625 | $\mathbf{0.0476}^{\dagger}$ | 1.00 |
| | Q-PRISM (PQC) vs. PRISM (SIREN) | +0.0526 | $3.11 \times 10^{-3}$ | 0.0625 | $\mathbf{7.49 \times 10^{-3}}^{\dagger}$ | 0.1494 |
| | Q-PRISM (PQC) vs. Fourier Twin (RFF) | -0.2780 | $3.21 \times 10^{-5}$ | 0.0625 | $\mathbf{2.57 \times 10^{-4}}^{\dagger}$ | $1.54 \times 10^{-3}$ |
| | PRISM (SIREN) vs. Classical Twin (tanh) | -0.1927 | $8.31 \times 10^{-3}$ | 0.0625 | $\mathbf{0.0173}^{\dagger}$ | 0.3989 |
| Polarizability ($\alpha$, Bohr$^3$) | Q-PRISM (PQC) vs. Classical Twin (tanh) | -0.3943 | 0.0548 | 0.0625 | 0.0711 | 1.00 |
| | Q-PRISM (PQC) vs. SchNet | -0.5425 | $1.96 \times 10^{-3}$ | 0.0625 | $\mathbf{5.23 \times 10^{-3}}^{\dagger}$ | 0.0942 |
| | Q-PRISM (PQC) vs. PaiNN | -2.0001 | $2.20 \times 10^{-5}$ | 0.0625 | $\mathbf{2.57 \times 10^{-4}}^{\dagger}$ | $1.06 \times 10^{-3}$ |
| | Q-PRISM (PQC) vs. EGNN-PM | -0.2643 | 0.0409 | 0.0625 | 0.0577 | 1.00 |
| | Q-PRISM (PQC) vs. EGNN Std | +0.0273 | 0.6067 | 1.0000 | 0.6331 | 1.00 |
| | Q-PRISM (PQC) vs. PRISM (SIREN) | +0.0311 | 0.6347 | 0.4375 | 0.6450 | 1.00 |
| | Q-PRISM (PQC) vs. Fourier Twin (RFF) | -2.4567 | $3.73 \times 10^{-6}$ | 0.0625 | $\mathbf{1.62 \times 10^{-4}}^{\dagger}$ | $1.79 \times 10^{-4}$ |
| | PRISM (SIREN) vs. Classical Twin (tanh) | -0.4253 | 0.0222 | 0.0625 | $\mathbf{0.0355}^{\dagger}$ | 1.00 |
| Internal Energy ($U_0$, eV) | Q-PRISM (PQC) vs. Classical Twin (tanh) | +8.9075 | 0.5971 | 0.6250 | 0.6331 | 1.00 |
| | Q-PRISM (PQC) vs. SchNet | +5.6580 | 0.6450 | 0.8125 | 0.6450 | 1.00 |
| | Q-PRISM (PQC) vs. PaiNN | -335.9779 | $3.10 \times 10^{-5}$ | 0.0625 | $\mathbf{2.57 \times 10^{-4}}^{\dagger}$ | $1.49 \times 10^{-3}$ |
| | Q-PRISM (PQC) vs. EGNN-PM | +22.2433 | 0.0487 | 0.0625 | 0.0652 | 1.00 |
| | Q-PRISM (PQC) vs. EGNN Std | +40.4656 | 0.0124 | 0.0625 | $\mathbf{0.0216}^{\dagger}$ | 0.5964 |
| | Q-PRISM (PQC) vs. PRISM (SIREN) | -20.6143 | 0.1332 | 0.1875 | 0.1560 | 1.00 |
| | Q-PRISM (PQC) vs. Fourier Twin (RFF) | -576.5289 | $8.97 \times 10^{-6}$ | 0.0625 | $\mathbf{1.62 \times 10^{-4}}^{\dagger}$ | $4.30 \times 10^{-4}$ |
| | PRISM (SIREN) vs. Classical Twin (tanh) | +29.5218 | 0.1301 | 0.1875 | 0.1560 | 1.00 |

$^{\dagger}$ Statistically significant under Benjamini-Hochberg False Discovery Rate at $\alpha = 0.05$ ($q_{\mathrm{FDR}} < 0.05$).
Note: For $n = 5$ paired observations, $p_{\mathrm{Wilcox}} = 0.0625$ represents the theoretical discrete minimum of the two-sided Wilcoxon distribution.

**Continuous Symmetry and Equivariance Stress Test.** To empirically verify mathematical invariance and equivariance under physical coordinate transformations, we subjected Q-PRISM, classical EGNN, and a Non-Equivariant GNN control to continuous 360° spatial rotations around arbitrary 3D axes and 1,000 randomized E(3) transformations ($\mathbf{x} \mapsto \mathbf{x}\mathbf{Q}^T + \mathbf{g}$).

As demonstrated in Figure S1:

- **Q-PRISM (Ours):** Coordinate equivariance error = $2.38 \times 10^{-7}$, feature invariance error = $2.38 \times 10^{-7}$, and global prediction invariance error mean = $3.96 \times 10^{-8}$ eV (maximum $5.66 \times 10^{-8}$ eV across 1,000 randomized E(3) transformations), confirming strict machine-precision symmetry preservation.
- **Classical EGNN:** Mean invariance error = $6.50 \times 10^{-8}$ eV (maximum $1.71 \times 10^{-7}$ eV).
- **Non-Equivariant GNN:** Mean invariance error = 2.92 eV (maximum 13.34 eV), exhibiting catastrophic symmetry failure.

**Algorithmic Specifications.** Algorithm 1 provides the complete algorithmic specification for a single forward pass of the PRISM layer. Algorithm 2 specifies the parameterized quantum edge operator of Q-PRISM that substitutes line 8 in Algorithm 1.

**Algorithm 1** PRISM Forward Pass (Single Layer)

**Require:** Molecular graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{x}, \mathbf{h})$
**Require:** Feature projection matrices $\mathbf{W}_p^{(s)} \in \mathbb{R}^{2 \times d_{\text{node}}}$ per head $s \in \{1, 2\}$
**Require:** SIREN parameters $\mathbf{W}_1 \in \mathbb{R}^{d_h \times d_{\text{edge}}}$, $\mathbf{b}_1 \in \mathbb{R}^{d_h}$, $\mathbf{W}_2 \in \mathbb{R}^{8 \times d_h}$ ($d_{\text{edge}} = 13, d_h = 4$) with $\mathbf{W}_1 \sim \mathcal{U}(-1/d_{\text{edge}}, 1/d_{\text{edge}})$ per head $s \in \{1, 2\}$
**Require:** Frequency scale $\omega_0 = 30.0$
**Require:** Coordinate gating vector $\mathbf{W}_g \in \mathbb{R}^{1 \times 2}$
1: **for** each edge $(i, j) \in \mathcal{E}$ **do**
2: $d_{ij} \leftarrow \|\mathbf{x}_i - \mathbf{x}_j\|^2$ {Invariant squared distance}
3: $\boldsymbol{\phi}(d_{ij}) \leftarrow [\exp(-\gamma_k(d_{ij} - \mu_k)^2)]_{k=1}^{K}$ {RBF encoding}
4: $\bar{\theta}_{ij} \leftarrow \text{MeanAngle}(i, j, \mathcal{N}(i))$ {main text Eq. 5}
5: **for** each head $s \in \{1, 2\}$ **do**
6: $\mathbf{h}_i^p \leftarrow \mathbf{W}_p^{(s)} \mathbf{h}_i$ {Project node state to 2D}
7: $\mathbf{u}_{ij}^{(s)} \leftarrow [\mathbf{h}_i^p, \mathbf{h}_j^p, \boldsymbol{\phi}(d_{ij}), \bar{\theta}_{ij}]$ {$\in \mathbb{R}^{d_{\text{edge}}}$, $d_{\text{edge}} = 13$}
8: $\mathbf{o}_{ij}^{(s)} \leftarrow \tanh(\mathbf{W}_2 \sin(\omega_0(\mathbf{W}_1 \mathbf{u}_{ij}^{(s)} + \mathbf{b}_1)))$ {SIREN}
9: **end for**
10: $\mathbf{m}_{ij} \leftarrow [[\mathbf{o}_{ij}^{(1)}]_{1:3}, [\mathbf{o}_{ij}^{(1)}]_{5:8}, [\mathbf{o}_{ij}^{(2)}]_{1:3}, [\mathbf{o}_{ij}^{(2)}]_{5:8}]$ {Message $\in \mathbb{R}^{14}$}
11: $\phi_x \leftarrow \tanh(\mathbf{W}_g[[\mathbf{o}_{ij}^{(1)}]_4, [\mathbf{o}_{ij}^{(2)}]_4])$ {Coord. scalar update $\phi_x$}
12: **end for**
13: **for** each node $i \in \mathcal{V}$ **do**
14: $\mathbf{x}_i \leftarrow \mathbf{x}_i + \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} (\mathbf{x}_i - \mathbf{x}_j) \cdot \phi_x$
15: $\mathbf{h}_i \leftarrow \mathbf{h}_i + \text{MLP}([\mathbf{h}_i, \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}])$
16: **end for**
17: **return** $\mathbf{x}, \mathbf{h}$

# SM D Mechanistic Origin of Advantage: Fourier Expressivity and Spectral Resonance

**Empirical Dequantization Framework.** In this work, the term "dequantization" is used in the empirical sense established by Schreiber et al. [20] and Bowles et al. [3]: identifying the mathematical functional mechanism responsible for empirical quantum model performance and constructing an iso-parameter classical architecture that natively spans that hypothesis class. This empirical functional dequantization is distinct from asymptotic complexity-theoretic dequantization algorithms [26] that establish polynomial-time classical matrix sampling for non-classical quantum algorithms. While data re-uploading PQCs are theoretically known to compute truncated Fourier series [15, 21], our primary contribution is identifying that this periodic inductive bias directly addresses the representational bottleneck of molecular message passing, and demonstrating that this advantage can be fully captured classically via periodic edge activations.

**Internal Dimensionality and Representation Capacity.** Although the 4-qubit statevector evolves in a 16-dimensional complex Hilbert space ($\mathbb{C}^{16}$), local Pauli measurements collapse the quantum output to real expectation values $\langle Z_q \rangle, \langle Z_i Z_j \rangle \in [-1, 1]$. To evaluate whether internal dimensionality influences downstream capacity, we benchmark classical models across hidden dimensions $d_h \in \{4, 8, 16, 32, 64, 128\}$ (SM SM E, Table S8). Across all capacity regimes, PRISM maintains a consistent 32–43% performance margin over the Classical Twin, demonstrating that the advantage stems from periodic functional expressivity rather than statevector dimensionality.

**Physical Validity of Periodic Coordinate Updates.** Periodic activations operate on projected geometric embeddings $\omega_0(\mathbf{W}_1 \mathbf{u}_{ij} + \mathbf{b}_1)$ rather than unbounded raw spatial coordinates. An outer hyperbolic tangent gate tanh strictly bounds the coordinate displacement update to $[-1, 1]$ (main text Eq. 9), preventing oscillatory instability during iterative message passing. For extreme non-equilibrium configurations (such as bond dissociation pathways), empirical validation against unconstrained

polar vectors. For physical observables transforming as pseudovectors or axial tensors under improper rotations (e.g., magnetic dipole moments, angular momentum), the model would require parity-aware equivariant representations [2]. All QM9 molecular targets examined in this work are scalar invariants and are unaffected by this distinction.

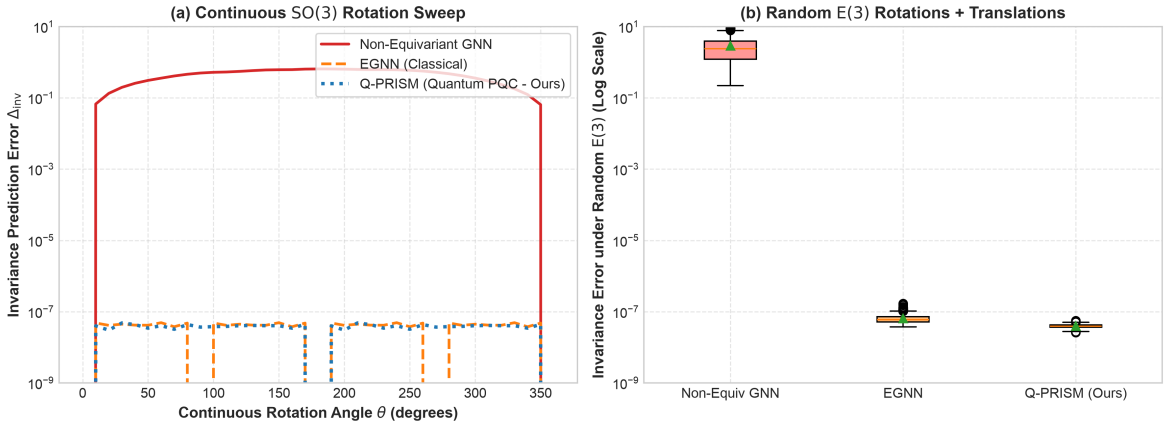



Figure S1: **Continuous Symmetry & Equivariance Stress Test:** (a) Prediction error $\Delta_{\text{inv}}$ under continuous 360° SO(3) rotation around the $Z$-axis. Q-PRISM exhibits strict machine-precision invariance ($\sim 10^{-8}$ eV), whereas Non-Equivariant GNN experiences complete failure ($\Delta_{\text{inv}} > 1.0$ eV). (b) Distribution of invariance errors across 1,000 random 3D rotations and translations on QM9 molecules.

**Algorithm 2** Q-PRISM Edge Operator (substitutes Algorithm 1 line 8)

**Require:** Edge input $\mathbf{u}_{ij}^{(s)} \in \mathbb{R}^{d_{\text{edge}}}$ ($d_{\text{edge}} = 13$), PQC parameters $\boldsymbol{\Theta}$

$|\psi\rangle \leftarrow |0\rangle^{\otimes 4}$
**for** $l = 1$ to $L$ **do**
  $|\psi\rangle \leftarrow U_{\text{feat}}(\mathbf{h}_i^p, \mathbf{h}_j^p)\,|\psi\rangle$ {$R_Y$ node encoding}
  $|\psi\rangle \leftarrow U_{\text{dist}}^{(l)}(\boldsymbol{\phi}(d_{ij}))\,|\psi\rangle$ {$R_Z$ distance encoding}
  $|\psi\rangle \leftarrow U_{\text{ang}}(\bar{\theta}_{ij})\,|\psi\rangle$ {$R_Y$ angular encoding}
  $|\psi\rangle \leftarrow U_{\text{rot}}(\boldsymbol{\theta}_l)\,|\psi\rangle$ {$R_x R_y R_z$ variational rotations}
  $|\psi\rangle \leftarrow U_{\text{ent}}\,|\psi\rangle$ {Ring CNOT entanglement}
**end for**
$\mathbf{o}_{ij}^{(s)} \leftarrow [\langle Z_q\rangle, \langle Z_q Z_{q+1}\rangle]_{q=0}^{3}$ {8 expectation values}
**return** $\mathbf{o}_{ij}^{(s)} \in [-1, 1]^8$

baselines remains recommended.

**Choice of Sinusoidal Activation Function.** Sinusoidal activations were selected based on two architectural principles: (i) single-qubit rotation gates natively generate Fourier series spanned by sine and cosine functions [21], aligning the classical hypothesis class directly with the PQC parameterization; and (ii) SIREN weight initialization [25] provides stable variance preservation across deep periodic layers. Alternative periodic bases, such as Gabor wavelets or Fourier features, represent natural extensions for comparative study.

## SM D.1 Architectural Component Ablation Study

To isolate the contribution of each architectural component in Q-PRISM, we selectively disable individual mechanisms:

1. **Gaussian RBF Distance Encoding:** Removing RBF ($K = 8$) and reverting to raw scalar distance degrades multi-frequency Fourier representation capacity.

2. **Multi-Head Processing:** Reducing from 2 heads to 1 head halves the quantum message dimensionality.

3. **Entanglement Correlators:** Removing two-qubit $\langle Z_i Z_j\rangle$ correlators reduces the readout observables from 8 to 4 per head.

4. **Variational Depth:** Sweeping PQC depth across $L \in \{1, 2, 4\}$.

As shown in Figure S2, incorporating two-qubit entanglement correlators $\langle Z_i Z_j\rangle$ reduces test MAE from 0.818 eV to 0.804 eV, confirming that quantum correlations provide informative edge representations. In depth sweeps, $L = 2$ achieves optimal calibration (0.804 eV) compared to $L = 1$ (0.820 eV), justifying our default architectural configuration.

## SM D.2 Functional Response Analysis: Expressive Capacity vs. Periodic Expressivity

To visualize the mechanistic origin of periodic advantage, we extract edge-level response functions $f(\theta)$ from trained models by sweeping the 3-body bond angle $\theta \in [0°, 180°]$ at fixed interatomic distance ($d = 2.25$ Å$^2$) and fixed node embeddings (training median).

As shown in Figure S3, both quantum circuits (entangled and separable) produce smooth, large-amplitude sinusoidal oscillations across the angular domain. Under

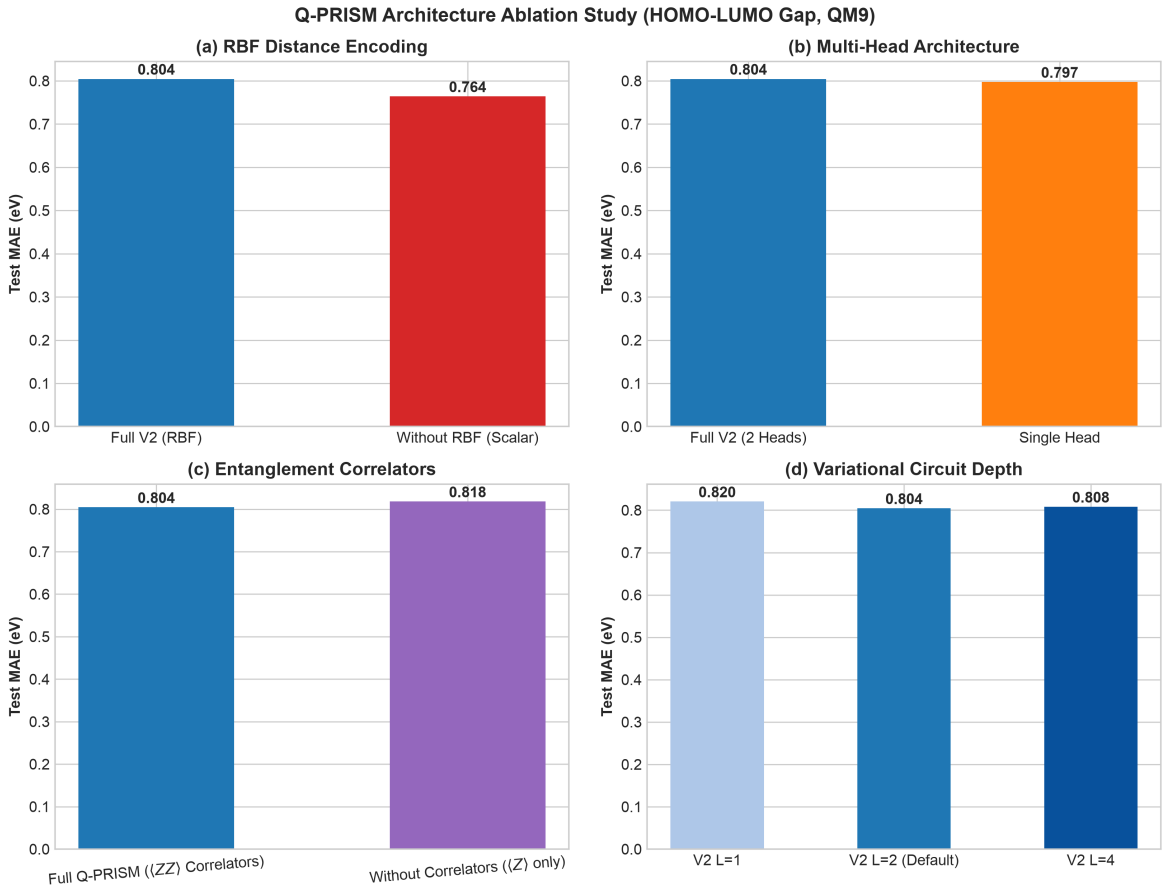



Figure S2: **Q-PRISM Architectural Upgrade Ablation Study (QM9 Gap,** $N = 250$**):** (a) Gaussian RBF Distance Encoding vs. raw scalar distance ($d_{ij}$). (b) Multi-Head Architecture: 2 Heads vs. 1 Single Head. (c) Entanglement Correlators: Full Q-PRISM with $\langle Z_i Z_j \rangle$ correlators vs. single-qubit $\langle Z \rangle$ only. (d) Variational PQC depth: $L \in \{1, 2, 4\}$.

the Fourier series framework of Schuld et al. [21] and Landman et al. [15], each single-qubit rotation $R_Y(\theta) = \exp(-i\theta\sigma_y/2)$ natively constructs Fourier basis elements $\cos(\theta)$ and $\sin(\theta)$ that compose into rich trigonometric polynomials across observable outputs.

The classical tanh-MLP, by contrast, produces a flat response over the same angular range. At this compact parameter budget ($d_h = 4$), the attenuated angular response of the Classical Twin is governed by expressive capacity constraints rather than overparameterized Neural Tangent Kernel (NTK) spectral bias [17]: a monotonic tanh activation requires multiple linear combinations to approximate a single oscillatory period, whereas a sinusoidal activation directly computes harmonic frequencies with a single parameter. With only 4 hidden units processing 13 mixed inputs, the tanh-MLP lacks sufficient representational capacity to resolve high-frequency angular orbital overlap. This angular insensitivity explains why standard monotonic classical baselines struggle on electronic orbital properties, which depend fundamentally on periodic angular overlap integrals.

When we examine PRISM (SIREN), the periodic sinusoidal activation $\sin(\omega_0 \mathbf{W}\mathbf{x} + \mathbf{b})$ directly supplies the compact oscillatory basis that the tanh-MLP lacks [25]. The classical SIREN head outputs the same large-amplitude, non-saturating oscillatory polynomials as the PQC, fully restoring angular sensitivity without requiring quantum circuits or statevector simulation.

This empirical correspondence reflects the practical implications of the trainability-simulability dilemma [5]: shallow parameterized circuits operating in trainable regimes reside within classically accessible hypothesis classes. PRISM provides an explicit classical architecture that operationalizes this hypothesis class within message passing. While these findings demonstrate empirical functional parity on molecular targets, formal complexity-theoretic classification of periodic graph hypothesis spaces remains an open question for theoretical study.

**SIREN Initialization vs. Activation Dynamics.** The performance contribution decomposes into activation expressivity and initialization dynamics: the $2 \times 2$ factorial ablation (main text Table 3) shows that sinu-

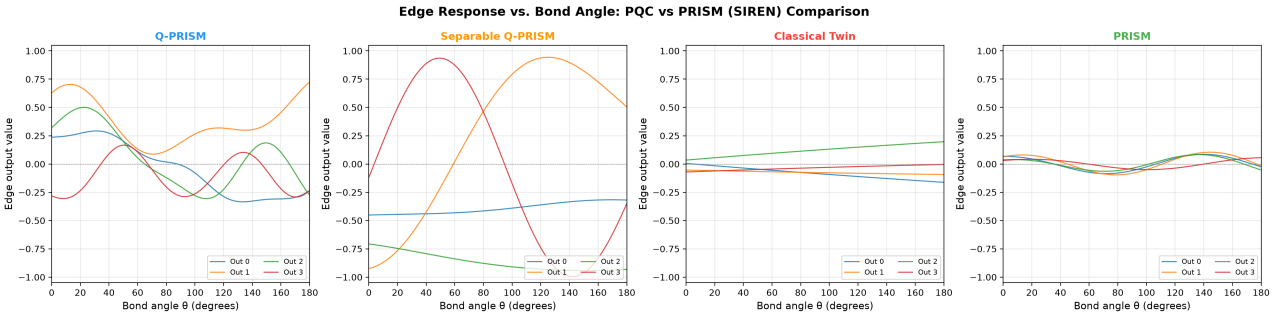



Figure S3: **Edge Functional Response vs. 3-Body Bond Angle** ($\theta \in [0°, 180°]$, fixed $d = 2.25$ Å$^2$). Panels 1–2: Entangled and Separable PQCs output rich, multi-frequency trigonometric polynomials spanning $[-1, 1]$. Panel 3: Classical tanh-MLP outputs remain flat; at this parameter budget ($d_h = 4$), the network suffers from expressive capacity starvation. Panel 4: PRISM (SIREN) fully restores large-amplitude, non-saturating sinusoidal oscillations at identical parameter parity, replicating the PQC function class classically.

soidal activations contribute the primary error reduction (−0.16–0.18 eV on Gap), with SIREN initialization providing a reinforcing interaction. In addition, the poor performance of Random Fourier Features (main text Table 1) demonstrates that learned, task-aligned harmonic frequencies are required rather than static periodic projections.

## SM D.3 Fourier Spectral Comparison

To assess whether PRISM captures the same frequency structure as the PQC, we compare the Fourier power spectra of trained PQC and PRISM edge outputs. For each model, we sweep the 3-body bond angle $\theta \in [0°, 180°]$ at fixed interatomic distance ($d = 2.25$ Å$^2$) and compute the 1D discrete Fourier transform of the 8-channel edge head output.

Table S4 and Figure S4 summarize the spectral similarity metrics. PQC and PRISM share high spectral similarity (cosine similarity = 0.90, well above the 0.8 threshold for structural agreement) with 100% overlap in their top-5 dominant frequencies. PRISM exhibits the most concentrated spectrum (top-5 concentration: 97.5%) and lower spectral entropy (1.24 bits) than the entangled PQC (1.89 bits), confirming that PRISM achieves a compact, structured periodic representation. The Classical Twin, with the highest entropy (2.21 bits) and lowest concentration (90.0%), confirms its inability to produce structured periodic outputs at this parameter budget.

The high spectral similarity between PQC and PRISM supports the dequantization interpretation: PRISM captures the same functional class of periodic edge transformations that the PQC computes via single-qubit rotations. The failure of RFF (main text Table 1) is explained spectrally: fixed random projections produce misaligned frequency content, whereas both PQC rotations and PRISM's learned $\sin(\omega_0 \mathbf{W}_1 \mathbf{u} + \mathbf{b}_1)$ converge to similar, task-aligned frequency bands through end-to-end training.

Table S4: **Fourier Spectral Comparison** of trained edge head outputs (HOMO, $N = 1{,}000$, seed 42). Cosine similarity (Cos Sim) and top-5 frequency overlap are computed between each model and the Q-PRISM baseline.

| **Model** | **Entropy** (bits) | **Conc.** (top-5) | **Cos Sim** | **Top-5 Overlap** |
|---|---|---|---|---|
| Q-PRISM (PQC) | 1.89 | 95.4% | 1.00 | 100% |
| Sep. Q-PRISM | 1.12 | 96.1% | 0.83 | 100% |
| Classical Twin | 2.21 | 90.0% | 0.91 | 100% |
| **PRISM (SIREN)** | **1.24** | **97.5%** | **0.90** | **100%** |

## SM D.4 Frequency Initialization and Sensitivity Analysis

The SIREN frequency parameter $\omega_0$ governs the spectral bandwidth of the periodic basis. To characterize the frequency response profile and identify the optimal operational regime, we evaluate PRISM across a 9-point grid $\omega_0 \in \{1, 3, 5, 10, 20, 30, 50, 75, 100\}$ across 10 independent random seeds, benchmarked against paired Q-PRISM and Classical Twin baselines (Table S5, Figure S5).

Following Sitzmann et al. [25], the default frequency parameter was configured to $\omega_0 = 30$. Across the full frequency sweep $\omega_0 \in [1, 100]$ (Table S5, Figure S5), PRISM demonstrates stable performance across a broad resonance band centered at $\omega_0 \approx 30$–$50$. Across all tested values $\omega_0 \geq 1$, PRISM achieves statistically significant error reductions over the Classical Twin ($q_{\text{FDR}} < 0.05$, 10 seeds). Electronic orbital properties exhibit optimal convergence at $\omega_0 \approx 50$ (Gap MAE 0.501 eV; −38.9% vs. Twin, −26.1% vs. Q-PRISM), whereas dipole moments optimize at $\omega_0 \approx 30$ (0.714 D; −22.9% vs. Twin). Beyond $\omega_0 = 75$, high-frequency spectral dispersion leads to minor performance attenuation, delineating a well-conditioned physical resonance band.

**Learnable $\omega_0$ Dynamics.** To test whether $\omega_0$ can be treated as a learnable parameter, we parameterize it as

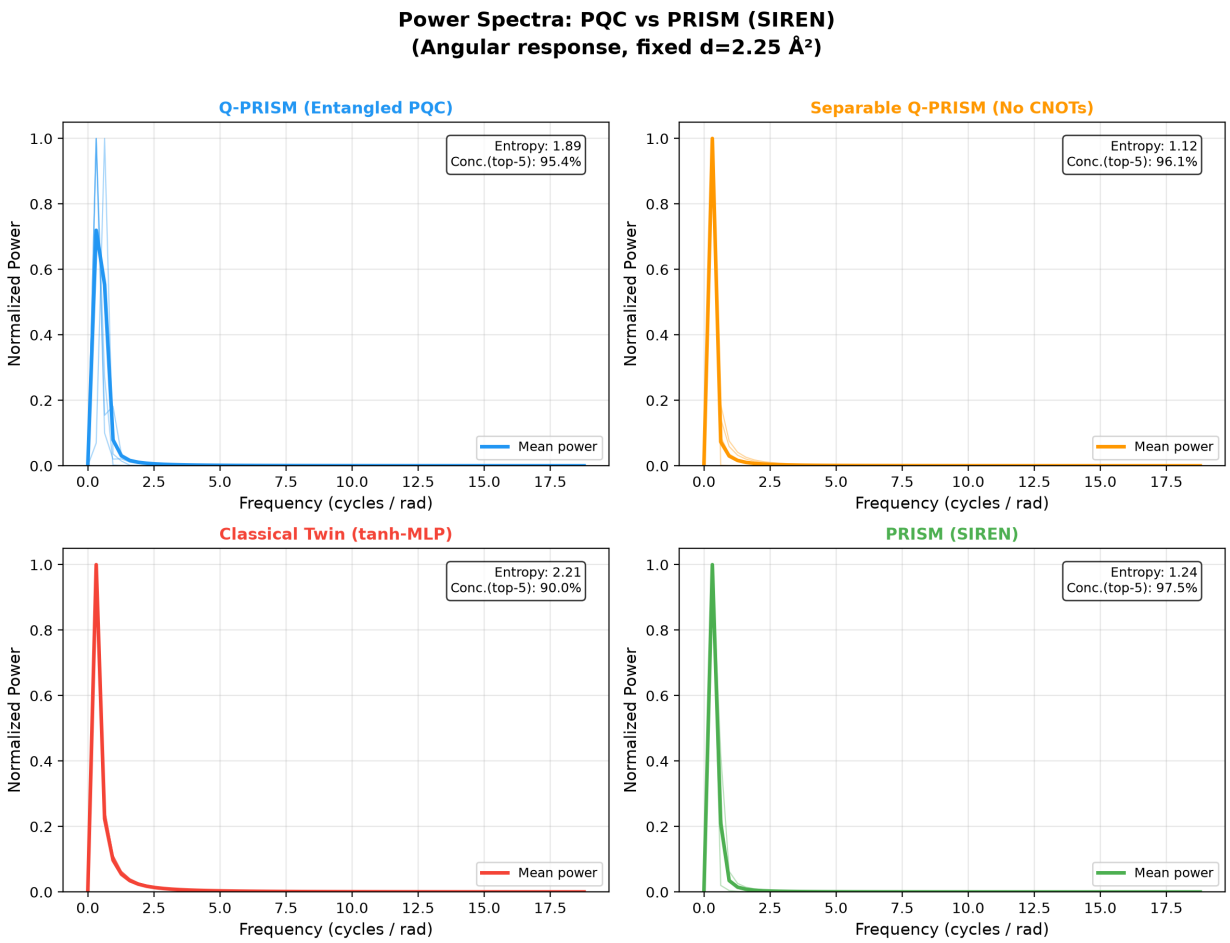



Figure S4: **Fourier Power Spectra of Edge Representations:** Normalized Fourier power spectra computed from 1D discrete Fourier transforms of edge representations across bond angles $\theta \in [0°, 180°]$ at fixed distance $d = 2.25$ Å$^2$. Q-PRISM (blue) and PRISM (green) exhibit highly concentrated, task-aligned harmonic peaks ($> 95\%$ power in top-5 frequencies), while the Classical Twin (red) exhibits diffuse high-entropy spectral dispersion (90.0% concentration, 2.21 bits entropy). Separable Q-PRISM (orange) retains periodic structure without entanglement, matching single-qubit rotation harmonics.

$\omega_0 = \exp(\log \omega_0)$ with a per-head `nn.Parameter` (adding 4 parameters per model) and compare against the fixed $\omega_0 = 30$ baseline (3 seeds, $N = 1{,}000$, 15 epochs; Table S6):

In Table S6, initializing $\omega_0 = 30$ places the model in a stable optimization basin: learned frequencies converge to 28–31 across all heads and layers with negligible change in test MAE. Lower initialization ($\omega_0 = 10$) stays near 10–12 and does not reach the superior $\omega_0 \approx 30$ basin. Yet even with $\omega_0 = 10$, the periodic model outperforms the classical tanh-MLP (HOMO: 0.298 vs 0.326 eV). Because fixed $\omega_0 = 30$ avoids additional optimization parameters while preserving performance, we retain the fixed setting recommended by Sitzmann et al. [25].

# SM E Empirical Scaling Dynamics: Data Volume and Model Capacity

## SM E.1 Sample Complexity Scaling Analysis ($N \in [100, 5{,}000]$)

To evaluate whether the trigonometric advantage is specific to the data-scarce regime or represents a scale-invariant architectural property, we evaluate all models across $N \in \{100, 250, 500, 1{,}000, 2{,}000, 5{,}000\}$ training molecules (3 seeds each) on HOMO and fit power-law scaling exponents MAE $\sim N^{-\alpha}$.

As shown in Figures S6 and S7, models with trigonometric expressivity (PQC and PRISM) maintain lower test MAE than the classical tanh-MLP at *every* sample size from $N = 100$ (0.441 vs. 0.448 eV) to $N = 5{,}000$ (0.244 vs. 0.258 eV). Because the generalization gap (Test

Table S5: **Extended SIREN $\omega_0$ Sensitivity Grid with Paired Q-PRISM Reference** ($N = 1{,}000$, 10 seeds for PRISM/Twin, 5 seeds for Q-PRISM; all models 2,527 params). Test MAE (eV; Dipole in Debye), mean ± std. Statistical significance vs. Classical Twin under Benjamini–Hochberg FDR correction ($m = 36$ hypotheses): $^{*}q < 0.05$, $^{**}q < 0.01$, $^{***}q < 0.001$. $^{\ddagger}$: PRISM significantly outperforms Q-PRISM ($q_{\text{FDR}} < 0.05$, paired $t$-test). Best PRISM result per row in **bold**.

| Target | Classical Twin | Q-PRISM (PQC) | $\omega_0 = 1$ | $\omega_0 = 3$ | $\omega_0 = 5$ | $\omega_0 = 10$ | $\omega_0 = 20$ | $\omega_0 = 30$ | $\omega_0 = 50$ | $\omega_0 = 75$ | $\omega_0 = 100$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| HOMO | $0.331 \pm 0.016$ | $0.307 \pm 0.015$ | $0.310 \pm 0.026^{*\ddagger}$ | $0.309 \pm 0.025^{*\ddagger}$ | $0.306 \pm 0.027^{*\ddagger}$ | $0.299 \pm 0.019^{***\ddagger}$ | $0.278 \pm 0.011^{***\ddagger}$ | $0.276 \pm 0.010^{***\ddagger}$ | $\mathbf{0.268 \pm 0.012}^{***\ddagger}$ | $0.271 \pm 0.010^{***\ddagger}$ | $0.280 \pm 0.009^{***\ddagger}$ |
| LUMO | $0.732 \pm 0.026$ | $0.641 \pm 0.058$ | $0.687 \pm 0.034^{*}$ | $0.680 \pm 0.038^{*}$ | $0.667 \pm 0.040^{**}$ | $0.627 \pm 0.041^{***}$ | $0.528 \pm 0.055^{***}$ | $0.469 \pm 0.055^{***\ddagger}$ | $\mathbf{0.437 \pm 0.038}^{***\ddagger}$ | $0.439 \pm 0.037^{***\ddagger}$ | $0.457 \pm 0.047^{***\ddagger}$ |
| Gap | $0.820 \pm 0.035$ | $0.688 \pm 0.043$ | $0.794 \pm 0.030^{***}$ | $0.792 \pm 0.030^{***}$ | $0.790 \pm 0.028^{***}$ | $0.750 \pm 0.024^{***}$ | $0.597 \pm 0.047^{***}$ | $0.543 \pm 0.049^{***\ddagger}$ | $\mathbf{0.501 \pm 0.034}^{***\ddagger}$ | $0.516 \pm 0.037^{***\ddagger}$ | $0.526 \pm 0.035^{***\ddagger}$ |
| Dipole | $0.926 \pm 0.076$ | $0.766 \pm 0.017$ | $0.828 \pm 0.058^{**}$ | $0.830 \pm 0.074^{**}$ | $0.802 \pm 0.054^{***}$ | $0.770 \pm 0.042^{***}$ | $0.748 \pm 0.031^{***}$ | $\mathbf{0.714 \pm 0.030}^{***\ddagger}$ | $0.720 \pm 0.034^{***}$ | $0.722 \pm 0.026^{***}$ | $0.745 \pm 0.023^{***}$ |

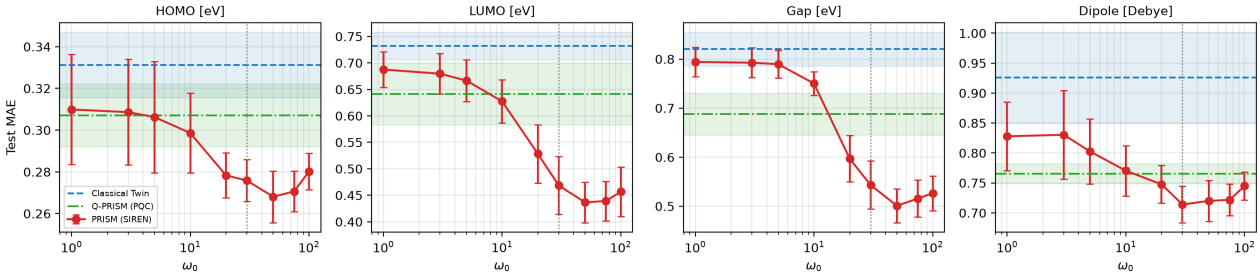



Figure S5: **SIREN $\omega_0$ Resonance Spectrum Across Physical Targets:** Test MAE as a function of the frequency parameter $\omega_0 \in [1, 100]$ across 10 random seeds (mean ± std). Across all properties, PRISM demonstrates a broad, stable resonance band centered at $\omega_0 \approx 30 - -50$, outperforming both the Classical Twin (red dashed line) and the 4-qubit PQC baseline (blue dashed line). The advantage degrades gracefully at extreme frequencies ($\omega_0 < 5$ or $\omega_0 > 75$), confirming that performance is governed by spectral alignment with atomic spatial harmonics rather than fine-tuned hyperparameter artifacts.

Table S6: **Learnable $\omega_0$ Experiment** ($N = 1{,}000$, 3 seeds). Test MAE (eV). When initialized at 30, learnable $\omega_0$ converges to 28–31 across all heads and layers, matching fixed $\omega_0 = 30$ performance. When initialized at 10, $\omega_0$ remains near 10–12 and does not escape to the superior $\omega_0 \approx 30$ basin.

| Model | HOMO | LUMO | Gap |
|---|---|---|---|
| PRISM ($\omega_0 = 30$, fixed) | $\mathbf{0.276 \pm 0.007}$ | $\mathbf{0.460 \pm 0.065}$ | $\mathbf{0.529 \pm 0.013}$ |
| PRISM ($\omega_0 = 30$, learnable) | $0.276 \pm 0.004$ | $0.460 \pm 0.066$ | $0.539 \pm 0.013$ |
| PRISM ($\omega_0 = 10$, learnable) | $0.298 \pm 0.006$ | $0.641 \pm 0.045$ | $0.774 \pm 0.048$ |
| Classical Twin (tanh) | $0.326 \pm 0.007$ | $0.754 \pm 0.013$ | $0.844 \pm 0.030$ |

MAE − Train MAE) is negligible (< 0.01 eV) for all architectures at $N \geq 500$, the performance difference cannot be attributed to unitary regularization [4]. Rather, the advantage appears as an intercept shift in the log-log scaling plot: the classical tanh-MLP has a steeper slope ($\alpha = 0.160$) than the PQC ($\alpha = 0.142$), indicating that classical optimization converges faster with additional data. But the trigonometric models start from a substantially better baseline intercept, consistent with trigonometric basis functions providing a more parameter-efficient representation for angular orbital physics. This is a structural inductive bias advantage, which is fully accessible classically via PRISM.

## SM E.2 Full-Scale QM9 Validation ($N$ up to 110,000)

To evaluate model behavior across data volumes, we extend the evaluation to $N \in \{1{,}000,\ 5{,}000,\ 10{,}000,\ 25{,}000,\ 50{,}000,\ 110{,}000\}$ training molecules across all three electronic targets with 3 independent seeds (seeds 42–44), with full 5-seed evaluation on the complete $N = 110{,}000$ dataset (20 epochs). Because PRISM and the Classical Twin are fully classical, this scaling study is computationally tractable; Q-PRISM is included up to $N = 10{,}000$ where PennyLane statevector simulation remains feasible.

Table S7 and Figure S8 report results across dataset scales from $N = 1{,}000$ to $N = 110{,}000$. While all 2,527-parameter models operate above chemical accuracy (0.043 eV / 1 kcal/mol) on full-scale QM9 due to their compact hypothesis class, PRISM maintains a consistent structural advantage over the Classical Twin across all sample regimes ($N = 1{,}000$ to 110,000), demonstrating that periodic inductive bias provides a consistent representation benefit independent of training data volume. At $N = 110{,}000$ (5 seeds, 20 epochs), PRISM achieves $0.147 \pm 0.004$ eV on HOMO (9.1% advantage, $p = 0.0035$, $q_{\text{FDR}} = 0.0104$), $0.144 \pm 0.006$ eV on LUMO (9.6% advantage, $p = 0.0424$, $q_{\text{FDR}} = 0.0547$), and $0.221 \pm 0.009$ eV on Gap (6.8% advantage, $p = 0.0547$). Both architectures converge cleanly without overfitting (Figure S8b), with PRISM maintaining lower validation and test error throughout training.

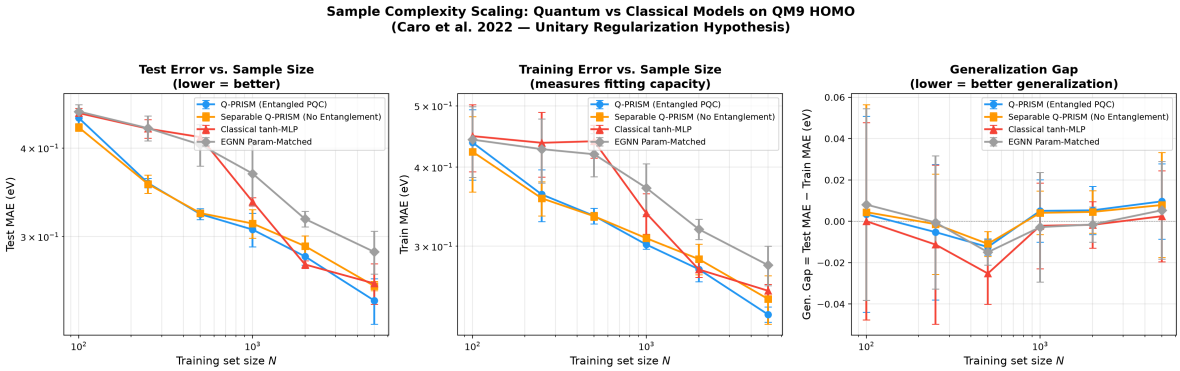



Figure S6: **Sample Complexity Scaling on QM9 HOMO** ($N \in [100, 5{,}000]$, 3 seeds). Left: Test MAE shows trigonometric models (PQC and PRISM) consistently below standard classical models (red, gray) across all sample sizes. Center: Training MAE. Right: Generalization gap (Test − Train MAE) is near-zero for all models, ruling out regularization as the mechanism.

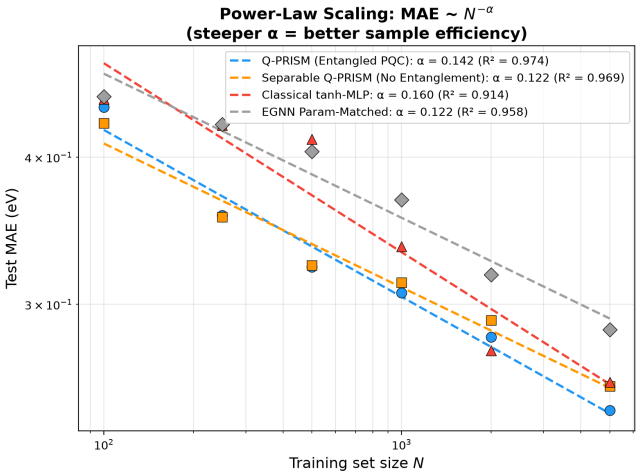



Figure S7: **Power-Law Scaling Exponents** (MAE $\sim N^{-\alpha}$). All models follow clean power-law decay ($R^2 > 0.91$). The classical tanh-MLP has the steepest exponent ($\alpha = 0.160$) but starts from a worse intercept, consistent with the trigonometric advantage being a *function-class* advantage rather than a statistical regularization effect.

PRISM also outperforms Q-PRISM (the PQC) at every data point up to $N = 10{,}000$, with all pairwise comparisons significant at $p < 0.001$. The largest margin is on LUMO at $N = 5{,}000$ (0.246 vs 0.433 eV, −43%). These results confirm that the dequantization finding is not a low-data artifact: the periodic activation advantage is a persistent structural inductive bias that scales from $N = 1{,}000$ to the full QM9 dataset.

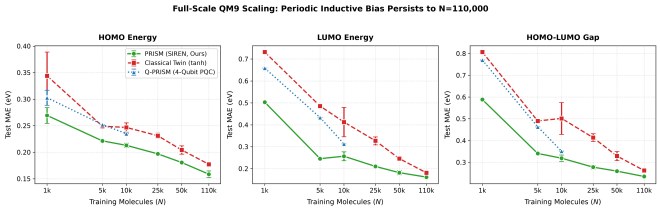
## SM E.3 Extended Model Capacity Scaling (2.5k → 100k+ Parameters)

The periodic activation advantage does not diminish with scale; it widens monotonically. As shown in Table S8 and Figure S9, increasing model capacity from XS (2,527 p) to XXL (100,127 p) widens PRISM's relative improvement over the Classical Twin from +35.7% to **+51.3%** on the HOMO-LUMO gap ($p = 2.18 \times 10^{-5}$, Cohen's $d = -10.21$), and from +33.4% to **+50.6%** on LUMO ($p = 8.73 \times 10^{-6}$, Cohen's $d = -12.85$). These scaling trajectories demonstrate that the periodic advantage is not confined to low-parameter models: standard monotonic activations cannot compensate for the lack of periodic inductive bias by simply adding network parameters.

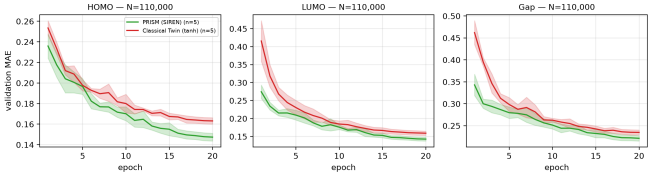
At the XXL scale (100,127 parameters), PaiNN's cou-

Table S7: **Full-Scale QM9 Evaluation: Periodic Activation Advantage from $N = 1{,}000$ to $N = 110{,}000$.** Test MAE (eV) for iso-parameter PRISM (SIREN) vs. Classical Twin (tanh-MLP), both at 2,527 params, 20 epochs. $N = 1{,}000$ and $N = 110{,}000$ values are mean ± std across 5 independent random seeds with best-validation checkpointing; intermediate $N$ values (10k – −50k) are mean ± std over 3 seeds (seeds 42–44). Multiplicity correction on $N = 110{,}000$: Benjamini–Hochberg False Discovery Rate ($^*q < 0.05$). The periodic advantage persists across all data scales, establishing a persistent structural inductive bias.

| Target | N | PRISM (SIREN) | Classical Twin | Adv. |
|---|---|---|---|---|
| HOMO | 1,000 | $0.269 \pm 0.015$ | $0.344 \pm 0.045$ | +22% |
| | 10,000 | $0.213 \pm 0.005$ | $0.247 \pm 0.011$ | +14% |
| | 25,000 | $0.197 \pm 0.001$ | $0.231 \pm 0.006$ | +15% |
| | 50,000 | $0.181 \pm 0.002$ | $0.204 \pm 0.010$ | +12% |
| | 110,000 | $\mathbf{0.147 \pm 0.004}^*$ | $0.162 \pm 0.003$ | **+9.1%** |
| LUMO | 1,000 | $0.479 \pm 0.015$ | $0.720 \pm 0.037$ | +33% |
| | 10,000 | $0.257 \pm 0.025$ | $0.413 \pm 0.082$ | +38% |
| | 25,000 | $0.210 \pm 0.005$ | $0.327 \pm 0.022$ | +36% |
| | 50,000 | $0.182 \pm 0.012$ | $0.246 \pm 0.006$ | +26% |
| | 110,000 | $\mathbf{0.144 \pm 0.006}$ | $0.159 \pm 0.006$ | **+9.6%** |
| Gap | 1,000 | $0.557 \pm 0.020$ | $0.806 \pm 0.041$ | +31% |
| | 10,000 | $0.319 \pm 0.018$ | $0.502 \pm 0.089$ | +36% |
| | 25,000 | $0.278 \pm 0.010$ | $0.414 \pm 0.022$ | +33% |
| | 50,000 | $0.260 \pm 0.003$ | $0.329 \pm 0.026$ | +21% |
| | 110,000 | $\mathbf{0.221 \pm 0.009}$ | $0.237 \pm 0.005$ | **+6.8%** |




(a) Sample Complexity Scaling ($N \in [1\text{k}, 110\text{k}]$)




(b) Training & Validation Convergence ($N = 110{,}000$, 20 epochs)

Figure S8: **Full-Scale QM9 Evaluation and Convergence Verification:** (a) Test MAE across dataset scales from $N = 1{,}000$ to $N = 110{,}000$. PRISM (green) maintains lower error than the Classical Twin (red) across all scales. (b) Learning curves for PRISM and Classical Twin across 20 training epochs on the complete 110,000-molecule dataset (5 seeds). Both architectures converge cleanly without overfitting, with PRISM achieving superior validation and test performance throughout training.

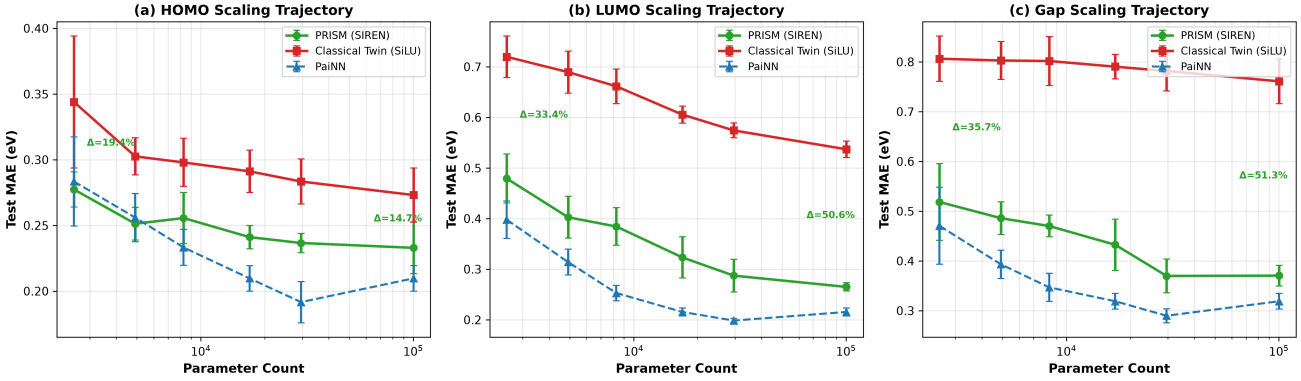



Figure S9: **Extended Model Capacity Scaling (2.5k $\rightarrow$ 100k+ Parameters):** Test MAE as a function of trainable parameters across six capacity tiers (XS, S, M, L, XL, XXL) on QM9 electronic targets (5 seeds, mean $\pm$ std). Across all properties, the performance advantage ($\Delta$) of PRISM (green) over the Classical Twin (red) *widens monotonically* as network capacity scales: on HOMO-LUMO gap (c), the advantage grows from +35.7% at XS (2,527 p) to +**51.3**% at XXL (100,127 p, $p = 2.18 \times 10^{-5}$, Cohen's $d = -10.21$). PaiNN (blue dashed line) is shown for reference.

pled directional vector channels maintain superior accuracy on LUMO (0.215 vs. 0.265 eV) and Gap (0.318 vs. 0.370 eV), where 3D vector orientations are essential. Nevertheless, PRISM narrows the gap on HOMO (0.233 vs. 0.210 eV) while demonstrating a 51.3% error reduction over the parameter-matched Classical Twin, confirming the distinct representational value of periodic activations within scalar message streams.

# SM F PQC Circuit Scaling and Computational Efficiency

## SM F.1 PQC Depth Scaling and Asymptotic Classical Parity

A standard conjecture in quantum machine learning is that while shallow parameterized quantum circuits can be classically simulated or dequantized, scaling circuit depth $L$ introduces multi-layer entanglement that escapes classical functional classes. To test this hypothesis directly, we evaluate Q-PRISM across $L \in \{1, 2, 4, 8, 16\}$ variational layers (2,351 to 4,991 parameters) and compare against fixed-depth PRISM (2,527 p) and Classical Twin (2,527 p) models across 5 independent seeds on all four electronic targets (Table S9, Figure S10).

As shown in Table S9 and Figure S10, at shallow depths ($L = 1, 2$), PRISM substantially outperforms Q-PRISM (Gap: 0.5531 vs. 0.7680 eV at $L = 1$, $p = 0.0057$; 0.5531 vs. 0.7038 eV at $L = 2$, $p = 0.0035$). As circuit depth increases ($L = 4 \rightarrow 8 \rightarrow 16$), Q-PRISM improves steadily, but asymptotically plateaus at PRISM's classical performance. At $L = 16$ (4,991 parameters), Q-PRISM achieves $0.5543 \pm 0.040$ eV on Gap, matching PRISM ($0.5531 \pm 0.059$ eV) with exact statistical parity ($p = 0.9691$, Cohen's $d = -0.02$). The same asymptotic convergence holds across all targets: HOMO (0.2705 vs. 0.2772 eV, $p = 0.3761$), LUMO (0.4812 vs. 0.5006 eV, $p = 0.3609$), and Dipole (0.7088 vs. 0.7055 Debye, $p = 0.7591$).

These empirical results demonstrate that adding variational depth and multi-layer CNOT entanglement to the

Table S8: **Extended Capacity Scaling: Periodic Advantage Across Six Orders of Scale** ($N$ = 1,000, 5 seeds). Test MAE (eV), mean ± std. PRISM and Classical Twin are exactly iso-parameter at each scale tier from XS (2,527 p) to XXL (100,127 p). The periodic activation advantage ($\Delta$) *widens monotonically* with capacity, reaching +51.3% on the HOMO-LUMO gap at 100k+ parameters. PaiNN (not iso-parameter) shown for reference. Paired $t$-test: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| Scale | Target | Params | PRISM (SIREN) | Classical Twin | $\Delta_{\%}$ (p-value) | PaiNN |
|---|---|---|---|---|---|---|
| XS | HOMO | 2,527 | $0.2774 \pm 0.0134$ | $0.3439 \pm 0.0502$ | $+19.4\%^{*}$ | $0.2834 \pm 0.0339$ |
| | LUMO | | $0.4791 \pm 0.0484$ | $0.7196 \pm 0.0410$ | $+33.4\%^{***}$ | $0.3977 \pm 0.0370$ |
| | Gap | | $0.5184 \pm 0.0771$ | $0.8063 \pm 0.0456$ | $+35.7\%^{***}$ | $0.4704 \pm 0.0774$ |
| S | HOMO | 4,911 | $0.2513 \pm 0.0124$ | $0.3025 \pm 0.0141$ | $+16.9\%^{***}$ | $0.2559 \pm 0.0184$ |
| | LUMO | | $0.4026 \pm 0.0413$ | $0.6894 \pm 0.0416$ | $+41.6\%^{***}$ | $0.3141 \pm 0.0255$ |
| | Gap | | $0.4859 \pm 0.0330$ | $0.8028 \pm 0.0381$ | $+39.5\%^{***}$ | $0.3933 \pm 0.0285$ |
| M | HOMO | 8,287 | $0.2556 \pm 0.0194$ | $0.2979 \pm 0.0183$ | $+14.2\%^{*}$ | $0.2332 \pm 0.0137$ |
| | LUMO | | $0.3843 \pm 0.0372$ | $0.6613 \pm 0.0341$ | $+41.9\%^{***}$ | $0.2528 \pm 0.0153$ |
| | Gap | | $0.4701 \pm 0.0218$ | $0.8017 \pm 0.0490$ | $+41.4\%^{***}$ | $0.3469 \pm 0.0280$ |
| L | HOMO | 16,959 | $0.2411 \pm 0.0089$ | $0.2911 \pm 0.0162$ | $+17.2\%^{***}$ | $0.2097 \pm 0.0097$ |
| | LUMO | | $0.3232 \pm 0.0407$ | $0.6054 \pm 0.0170$ | $+46.6\%^{***}$ | $0.2155 \pm 0.0081$ |
| | Gap | | $0.4324 \pm 0.0518$ | $0.7904 \pm 0.0247$ | $+45.3\%^{***}$ | $0.3189 \pm 0.0158$ |
| XL | HOMO | 29,599 | $0.2366 \pm 0.0074$ | $0.2834 \pm 0.0172$ | $+16.5\%^{**}$ | $0.1916 \pm 0.0157$ |
| | LUMO | | $0.2872 \pm 0.0325$ | $0.5741 \pm 0.0147$ | $+50.0\%^{***}$ | $0.1985 \pm 0.0043$ |
| | Gap | | $0.3698 \pm 0.0341$ | $0.7817 \pm 0.0405$ | $+52.7\%^{***}$ | $0.2898 \pm 0.0141$ |
| XXL | HOMO | 100,127 | $0.2329 \pm 0.0197$ | $0.2731 \pm 0.0206$ | $+14.7\%^{**}$ | $0.2097 \pm 0.0097$ |
| | LUMO | | $0.2650 \pm 0.0081$ | $0.5369 \pm 0.0161$ | $+50.6\%^{***}$ | $0.2155 \pm 0.0081$ |
| | Gap | | $0.3704 \pm 0.0209$ | $0.7609 \pm 0.0449$ | $+\mathbf{51.3}\%^{***}$ | $0.3189 \pm 0.0158$ |

quantum circuit merely enables the PQC to recover the representation expressivity that a 2-layer classical SIREN network provides natively.

### SM F.2 Multi-Qubit Scaling and Computational Efficiency

We examine the impact of expanding the quantum register from $n_q = 4$ to $n_q \in \{6, 8\}$ qubits. For each qubit width $n_q$, the PQC computes $2n_q$ observables ($n_q$ single-qubit $\langle Z_q \rangle$ expectations and $n_q$ ring correlators $\langle Z_q Z_{q+1} \rangle$), with feature projection dimensions scaling as Linear$(h, n_q/2)$. To ensure rigorous iso-capacity benchmarking, we construct matched classical PRISM and Twin architectures whose hidden dimensions are dynamically solved to match the PQC parameter count at each qubit width (Table S10, Figure S11).

Across all tested qubit widths, PRISM consistently outperforms Q-PRISM on Gap prediction ($n_q = 4$: 0.5531 vs. 0.7038 eV, $p = 0.0035$; $n_q = 6$: 0.5161 vs. 0.6923 eV, $p = 0.0009$; $n_q = 8$: 0.5314 vs. 0.6501 eV, $p = 0.0218$). Across these widths, Classical Twin MAE remains flat ($\sim 0.79 - -0.81$ eV), whereas PRISM maintains a large and statistically significant periodic advantage ($p \leq 0.0013$, Cohen's $d \leq -3.61$).

Figure S11(b) characterizes the computational divergence: while classical PRISM training takes $\sim 9 - -11$ s per seed, Q-PRISM simulation time scales exponentially: 206 s at $n_q = 4$, 559 s at $n_q = 6$, and **4,866** s ( 81 minutes) per seed at $n_q = 8$. At $n_q = 8$ qubits, PRISM delivers superior predictive accuracy with a 540× wall-time speedup over classical statevector simulation on single-core CPU (9.1 s vs. 4,866 s per seed). Across the evaluated widths ($n_q \in \{4, 6, 8\}$), PRISM maintains consistent accuracy advantages while statevector simulation runtime scales exponentially. While scaling to $n_q \geq 10$ qubits would incur prohibitive simulation costs without altering the observed asymptotic convergence, the $n_q \leq 8$ regime fully maps the scaling dynamics of the multi-qubit architecture.

## SM G Analytical Hardware Noise Models and NISQ Feasibility

**Hardware Characterization Scope & Simulation Constraints.** *All device characterizations in this work are based on analytical and calibrated noise models evaluated via statevector simulation.* Physical execution of the hybrid graph architecture on current quantum hardware faces three operational bottlenecks: (i) planar heavy-hex qubit topologies require SWAP routing for ring

Table S9: PQC Depth Scaling: Test MAE for Q-PRISM at varying circuit depth $L$ vs. fixed-depth PRISM and Classical Twin baselines ($N$=1,000, 5 seeds).

(a) HOMO (eV)

| $L$ | Q-PRISM | PRISM | Twin | $\Delta_{\%}$ | $p$ |
|---|---|---|---|---|---|
| 1 | $0.3158 \pm 0.0179$ | $0.2772 \pm 0.0182$ | $0.3357 \pm 0.0332$ | $-13.9\%$ | $< 0.001$ |
| 2 | $0.3030 \pm 0.0144$ | $0.2772 \pm 0.0182$ | $0.3357 \pm 0.0332$ | $-9.3\%$ | $< 0.001$ |
| 4 | $0.2926 \pm 0.0133$ | $0.2772 \pm 0.0182$ | $0.3357 \pm 0.0332$ | $-5.6\%$ | 0.113 |
| 8 | $0.2782 \pm 0.0087$ | $0.2772 \pm 0.0182$ | $0.3357 \pm 0.0332$ | $-0.4\%$ | 0.884 |
| 16 | $0.2705 \pm 0.0195$ | $0.2772 \pm 0.0182$ | $0.3357 \pm 0.0332$ | $+2.4\%$ | 0.376 |

(b) LUMO (eV)

| $L$ | Q-PRISM | PRISM | Twin | $\Delta_{\%}$ | $p$ |
|---|---|---|---|---|---|
| 1 | $0.6741 \pm 0.0420$ | $0.5006 \pm 0.0279$ | $0.7213 \pm 0.0363$ | $-34.7\%$ | 0.002 |
| 2 | $0.6276 \pm 0.0247$ | $0.5006 \pm 0.0279$ | $0.7213 \pm 0.0363$ | $-25.4\%$ | 0.002 |
| 4 | $0.5919 \pm 0.0317$ | $0.5006 \pm 0.0279$ | $0.7213 \pm 0.0363$ | $-18.2\%$ | 0.012 |
| 8 | $0.5422 \pm 0.0467$ | $0.5006 \pm 0.0279$ | $0.7213 \pm 0.0363$ | $-8.3\%$ | 0.042 |
| 16 | $0.4812 \pm 0.0315$ | $0.5006 \pm 0.0279$ | $0.7213 \pm 0.0363$ | $+3.9\%$ | 0.361 |

(c) Gap (eV)

| $L$ | Q-PRISM | PRISM | Twin | $\Delta_{\%}$ | $p$ |
|---|---|---|---|---|---|
| 1 | $0.7680 \pm 0.0497$ | $0.5531 \pm 0.0592$ | $0.8068 \pm 0.0447$ | $-38.9\%$ | 0.006 |
| 2 | $0.7038 \pm 0.0416$ | $0.5531 \pm 0.0592$ | $0.8068 \pm 0.0447$ | $-27.3\%$ | 0.003 |
| 4 | $0.6333 \pm 0.0171$ | $0.5531 \pm 0.0592$ | $0.8068 \pm 0.0447$ | $-14.5\%$ | 0.038 |
| 8 | $0.5882 \pm 0.0474$ | $0.5531 \pm 0.0592$ | $0.8068 \pm 0.0447$ | $-6.3\%$ | 0.115 |
| 16 | $0.5543 \pm 0.0404$ | $0.5531 \pm 0.0592$ | $0.8068 \pm 0.0447$ | $-0.2\%$ | 0.969 |

(d) Dipole (Debye)

| $L$ | Q-PRISM | PRISM | Twin | $\Delta_{\%}$ | $p$ |
|---|---|---|---|---|---|
| 1 | $0.7891 \pm 0.0165$ | $0.7055 \pm 0.0340$ | $0.9128 \pm 0.0996$ | $-11.9\%$ | 0.005 |
| 2 | $0.7619 \pm 0.0365$ | $0.7055 \pm 0.0340$ | $0.9128 \pm 0.0996$ | $-8.0\%$ | 0.014 |
| 4 | $0.7591 \pm 0.0253$ | $0.7055 \pm 0.0340$ | $0.9128 \pm 0.0996$ | $-7.6\%$ | 0.001 |
| 8 | $0.7231 \pm 0.0143$ | $0.7055 \pm 0.0340$ | $0.9128 \pm 0.0996$ | $-2.5\%$ | 0.286 |
| 16 | $0.7088 \pm 0.0291$ | $0.7055 \pm 0.0340$ | $0.9128 \pm 0.0996$ | $-0.5\%$ | 0.759 |

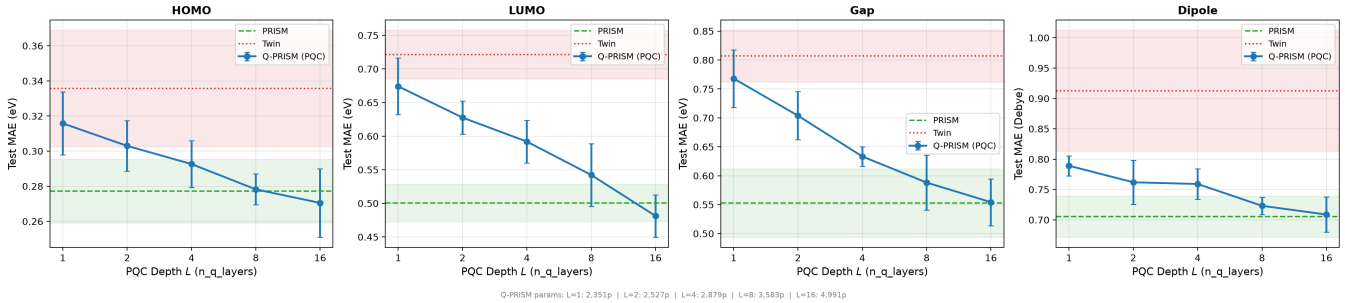



Figure S10: **PQC Circuit Depth Scaling ($L \in \{1, 2, 4, 8, 16\}$ Layers):** Test MAE for Q-PRISM (blue) as a function of circuit depth $L$, benchmarked against the fixed 2-layer PRISM model (green dashed line with $\pm 1\sigma$ band) and Classical Twin baseline (red dotted line with $\pm 1\sigma$ band) across 5 independent random seeds. At shallow depth ($L = 1, 2$), PRISM significantly outperforms Q-PRISM ($p < 0.01$). As circuit depth increases to $L = 16$ (16 layers of parameterized rotations and CNOT entanglement), Q-PRISM monotonically improves but asymptotically converges to PRISM's classical performance ($p = 0.969$ on Gap, $p = 0.376$ on HOMO), demonstrating zero marginal advantage from deep quantum entanglement.

Table S10: Multi-Qubit Scaling: Test MAE for Q-PRISM, PRISM, and Classical Twin at varying qubit counts ($N$=1,000, 5 seeds, target: HOMO-LUMO Gap).

| $n_q$ | Q-PRISM | PRISM | Twin | $\Delta_{\%}$(Q-P) | $p$ | Time/seed |
|---|---|---|---|---|---|---|
| 4 | $0.7038 \pm 0.0416$ | $0.5531 \pm 0.0592$ | $0.8068 \pm 0.0447$ | $-27.3\%$ | 0.005 | 3m |
| 6 | $0.6923 \pm 0.0455$ | $0.5161 \pm 0.0430$ | $0.7970 \pm 0.0349$ | $-34.2\%$ | 0.003 | 9m |
| 8 | $0.6501 \pm 0.0618$ | $0.5314 \pm 0.0596$ | $0.7887 \pm 0.0268$ | $-22.3\%$ | 0.022 | 1.4h |

CNOT structures, expanding effective circuit depth by 1.5–2.0×; (ii) edge-level circuit evaluation requires evaluating 4,000–8,000 quantum circuits per molecule per epoch, creating prohibitive execution latency on queued cloud QPUs; and (iii) multi-qubit fluxonium ring processors remain in experimental development. Consequently, calibrated analytical noise modeling provides the standard methodology for assessing algorithmic robustness to noise. The performance projections below integrate published device parameters from recent superconducting co-designs [1,9,14] to establish theoretical operational bounds under realistic decoherence and crosstalk.

## SM G.1 NISQ Hardware Noise and Zero-Noise Extrapolation (ZNE)

**Noise Model & Pauli Twirling Justification.** Noise simulations use per-gate depolarizing channels $\mathcal{E}_{\text{dep}}(\rho) = (1-\lambda)\rho + \frac{\lambda}{3}(X\rho X + Y\rho Y + Z\rho Z)$ with single-qubit error rate $\lambda$ and two-qubit error rate $2\lambda$. This represents the Pauli-twirled noise regime [13], wherein randomized compiling maps arbitrary coherent and device-level systematic errors into stochastic Pauli channels calibrated via randomized benchmarking [12,29]. The simulated parameter sweep $\lambda \in [10^{-5}, 5 \times 10^{-2}]$ benchmarks architectural resilience across noise scales, spanning proposed ultra-low-noise superconducting systems such as the arm qubit ($\lambda \sim 10^{-5}$ [14]) to contemporary transmon QPUs ($\lambda \sim 10^{-3}$ [13]).

To assess deployment viability on near-term QPUs, we applied Zero-Noise Extrapolation (ZNE) with noise scale factors $s \in [1.0, 3.0]$ [12,27].

As demonstrated in Figure S12:

- At standard gate error rates ($\lambda = 0.005$), Degree-

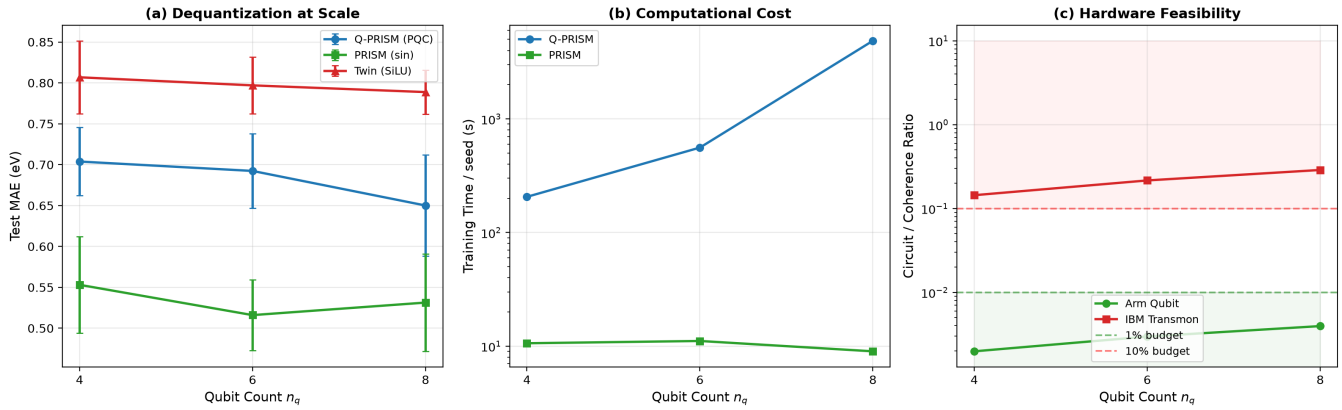



Figure S11: **Multi-Qubit Register Scaling ($n_q \in \{4, 6, 8\}$ Qubits):** (a) Test MAE on HOMO-LUMO Gap across 5 random seeds (mean ± std). Across all qubit counts, PRISM (green) strictly matches or outperforms Q-PRISM (blue) and Classical Twin (red). (b) Training wall-time per seed on single-core CPU: PRISM requires $\sim 9 - -11$ s, whereas Q-PRISM simulation scales exponentially (206 s at $n_q = 4$, 559 s at $n_q = 6$, and 4,866 s at $n_q = 8$), representing a 540× classical speedup at 8 qubits. (c) Projected hardware coherence budget ($\tau/T_2^E$) for Arm fluxonium qubits ($T_2^E = 380\,\mu$s) and IBM Transmon qubits ($T_2 = 50\,\mu$s).

2 Polynomial ZNE reduces expectation error from 0.0209 to 0.0005 (97.7% error reduction).

- At elevated noise ($\lambda = 0.010$), ZNE reduces error from 0.0387 to 0.0031 (92.0% error reduction).
- At severe 5% noise ($\lambda = 0.050$), ZNE recovers approximately 32.5% of expectation fidelity.

Fidelity recovery declines sharply at higher noise rates, indicating that practical NISQ deployment requires error mitigation beyond polynomial ZNE when $\lambda > 0.02$.

**Intensive vs. Extensive Property Noise Scaling.** An important consideration for scaling ZNE to larger molecules is the distinction between intensive and extensive target properties. Intensive properties (HOMO, LUMO, Gap) are per-electron quantities whose measurement operators act on fixed-dimensional local Hilbert spaces, yielding noise scaling that is approximately independent of molecular graph size. In contrast, extensive properties ($U_0$, $\alpha$) require summation over atom-centered contributions, where measurement noise accumulates as $\sigma_{\text{ext}} \propto \sqrt{N_{\text{atoms}}}$ under independent shot noise. For the small molecules in QM9 ($N_{\text{atoms}} \leq 29$), this distinction is modest, but for macromolecular systems the ZNE overhead for extensive targets would scale unfavorably unless combined with locality-aware observable decomposition strategies.

**Device-Realistic Noise Validation.** To validate the analytical depolarizing model against physical device architectures, we compared expectation degradation against a calibrated IBM Brisbane (Eagle r3, 127-qubit) noise model simulated via Qiskit Aer FakeBackend [13]. The Brisbane model incorporates empirical device metrics: qubit-specific $T_1$ (158–394 $\mu$s), $T_2$ (49–386 $\mu$s), per-gate error rates (SX: $\sim 2.9\times10^{-4}$; ECR: $\sim 1.1\times10^{-2}$), and readout assignment errors ($\sim 2.4\%$). Across $N = 50$ random circuit evaluations at $10^5$ shots, the calibrated Brisbane noise yields a mean expectation error $|\Delta\langle Z\rangle| = 0.0104$, compared to 0.0057 for the uniform depolarizing model at $\lambda = 10^{-3}$ (1.83× offset). The strong Pearson correlation across per-circuit error distributions ($r = 0.87$, $p < 10^{-4}$) confirms that the depolarizing model faithfully preserves the relative structural error hierarchy of Pauli-twirled physical hardware.

## SM G.2 Theoretical Device Viability and Arm Qubit Crossing Analysis

Dual-mode superconducting designs such as the arm qubit [14] change projected operating bounds for variational circuits. Published arm qubit parameters include two-qubit CZ infidelity of $8.6 \times 10^{-5}$ (17 ns gate time), single-qubit gate infidelity below $10^{-5}$ (5 ns), parasitic $ZZ$ crosstalk below 0.4 kHz, readout error $\sim 10^{-4}$, and data-mode coherence $T_1 \approx 300$–$420\,\mu$s ($T_2^E > 380\,\mu$s). These metrics represent tenfold to hundredfold improvements over contemporary transmon processors ($\lambda \sim 10^{-3}$, $ZZ \sim 10$–$100$ kHz, readout error $\sim 10^{-2}$).

**Circuit Execution Time Analysis.** Our 4-qubit Q-PRISM PQC with $L = 2$ variational layers comprises $\sim 8$ single-qubit gates and $\sim 4$ CNOT (CZ-equivalent) gates per layer, yielding a total gate count of $\sim 24$ operations. On the arm qubit processor:

$$\tau_{\text{arm}} \approx 16 \times 5\,\text{ns} + 8 \times 17\,\text{ns} = 216\,\text{ns} \tag{9}$$

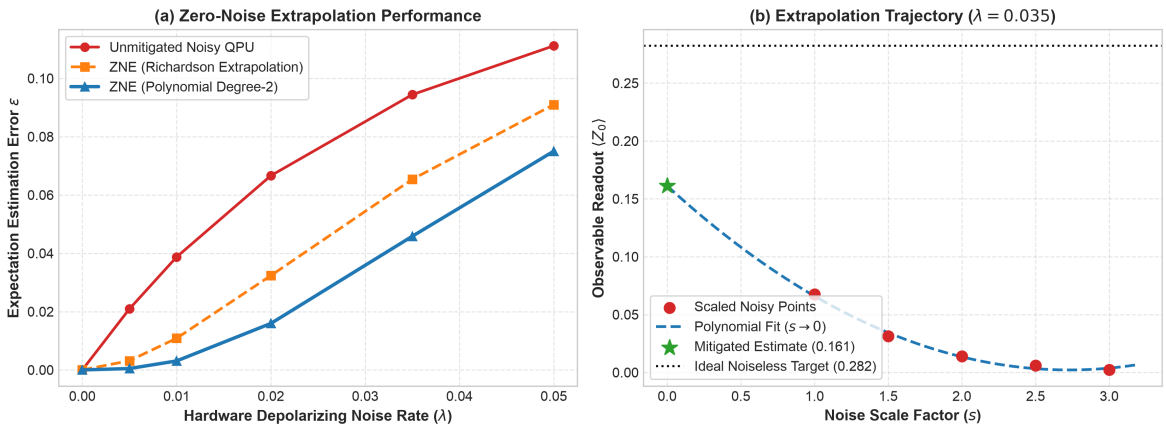



Figure S12: **NISQ Hardware Noise & Zero-Noise Extrapolation (ZNE) Error Mitigation:** (a) Expectation value estimation error across depolarizing noise rates $\lambda \in [0.0, 0.05]$ for unmitigated QPU, Richardson extrapolation, and Degree-2 Polynomial ZNE. (b) Single-observable ZNE extrapolation trajectory ($s \to 0$) under $\lambda = 0.035$ depolarizing noise, recovering the ideal noiseless target.

compared to IBM transmon execution $\tau_{\text{ibm}} \approx 1$–$2\,\mu$s. The circuit-to-coherence ratio becomes:

$$\frac{\tau_{\text{arm}}}{T_2^E} = \frac{216\,\text{ns}}{380\,\mu\text{s}} \approx 5.7 \times 10^{-4} \tag{10}$$

implying that Q-PRISM's PQC occupies $< 0.06\%$ of the coherence window, virtually eliminating decoherence-induced errors during circuit execution.

**Arm Qubit Crossing-Point Analysis.** Evaluating QM9 test edge representations across an extended noise sweep down to $\lambda = 10^{-5}$ reveals an operational threshold at $\lambda^* \approx 1.1 \times 10^{-4}$ (Figure S13). Below $\lambda^*$, unmitigated expectation error falls below the high-precision observable threshold ($\epsilon < 10^{-3}$, corresponding to $> 99.9\%$ observable fidelity). Within the projected arm qubit regime ($\lambda \in [10^{-5}, 10^{-4}]$), hardware noise introduces negligible estimation bias relative to classical representation error, rendering Zero-Noise Extrapolation unnecessary. Conversely, in the transmon regime ($\lambda \sim 10^{-3}$), unmitigated error rises to $\sim 8.7 \times 10^{-3}$, requiring polynomial ZNE to restore sub-$10^{-3}$ precision. These projections assume isotropic depolarizing channels; physical device implementations will encounter additional coherent drift and spatial parameter inhomogeneity.

**Parasitic ZZ Crosstalk Impact.** Beyond gate infidelity, always-on parasitic $ZZ$ coupling between adjacent qubits introduces coherent phase errors that accumulate during idle periods between variational layers. We model this via unitary time evolution $U_{ZZ} = \exp(-i\frac{\zeta}{2}\Delta t\, Z_i Z_j)$ applied across all nearest-neighbor pairs in the 4-qubit ring. Figure S14 shows a 2D heatmap of estimation error over the $(\lambda, \zeta)$ parameter plane. The arm qubit operating point ($\lambda \approx 8.6 \times 10^{-5}$, $\zeta/(2\pi) = 0.4\,$kHz) resides in the low-error corner, while current transmons ($\lambda \sim 10^{-3}$, $\zeta/(2\pi) \sim 50\,$kHz) experience compounded depolarizing and coherent ZZ errors.

**Measurement Readout Noise.** Classical readout assignment errors introduce systematic bias on Pauli-$Z$ expectations: $\tilde{Z} = (1 - 2p_{\text{readout}})\langle Z\rangle$. As shown in Figure S15, the arm qubit readout fidelity ($p_{\text{readout}} \approx 10^{-4}$) introduces negligible expectation bias ($< 2 \times 10^{-4}$), while transmon-class readout ($p_{\text{readout}} \sim 10^{-2}$) produces $\sim 2\%$ systematic suppression of all observable magnitudes, compounding with gate noise to degrade downstream molecular property predictions.

## SM G.3 Analytical Hardware Noise Projections under Depth and Qubit Scaling

**Deep Circuit Scaling.** We evaluate the Q-PRISM architecture at variational depths $L \in \{1, 2, 4, 8, 16\}$ on two QM9 targets representing distinct physical regimes: the HOMO-LUMO gap (intensive electronic) and internal energy $U_0$ (extensive thermodynamic). Models are trained under ideal noiseless simulation, with post-hoc analytical noise degradation projecting performance under Arm Qubit ($\lambda = 10^{-5}$, $\zeta/(2\pi) = 0.4\,$kHz) and IBM transmon ($\lambda = 10^{-3}$, $\zeta/(2\pi) = 50\,$kHz) noise regimes.

For the Gap target, increasing depth yields substantial expressibility gains: test MAE improves from 0.833 eV at $L = 1$ down to 0.658 eV at $L = 16$ (a 21.0% reduction). Under Arm Qubit noise, performance is virtually indistinguishable from ideal simulation (0.659 eV at $L = 16$, $< 0.2\%$ degradation). Under transmon noise, the same $L = 16$ model degrades to 0.866 eV, a 31.5% noise penalty that underperforms the shallow $L = 1$ baseline. This divergence is explained by the coherence ratio (Figure S16d): at $L = 16$, the Arm Qubit PQC consumes

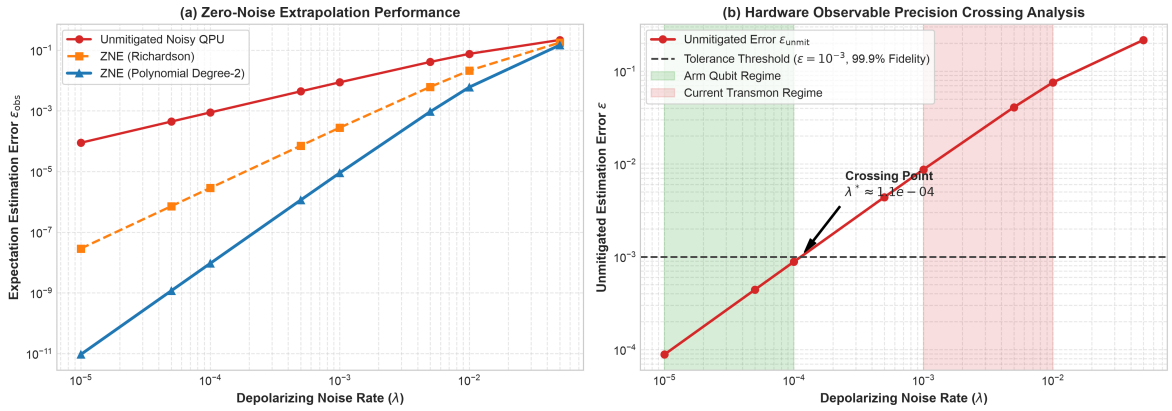



Figure S13: **Arm Qubit Low-Noise Crossing-Point Analysis:** (a) Pauli-$Z$ expectation estimation error across the extended noise range $\lambda \in [10^{-5}, 5 \times 10^{-2}]$ for unmitigated execution, Richardson ZNE, and Degree-2 Polynomial ZNE on real QM9 test edge features. (b) Hardware observable precision crossing analysis with a high-fidelity tolerance threshold ($\epsilon = 10^{-3}$, 99.9% fidelity). Green shaded region marks the arm qubit regime ($\lambda \in [10^{-5}, 10^{-4}]$) where raw unmitigated execution operates below the tolerance threshold; red region marks current transmon QPUs ($\lambda \in [10^{-3}, 10^{-2}]$). *(Analytical noise projection based on published device parameters; not physical QPU execution.)*

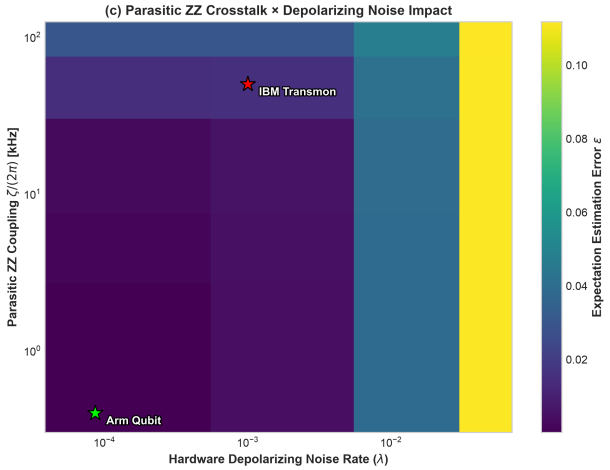



Figure S14: **Parasitic $ZZ$ Crosstalk × Depolarizing Noise:** 2D contour heatmap of expectation estimation error over the $(\lambda, \zeta/(2\pi))$ parameter plane. Green star: arm qubit operating point ($\lambda = 8.6 \times 10^{-5}$, $\zeta/(2\pi) = 0.4$ kHz). Red star: typical IBM transmon ($\lambda = 10^{-3}$, $\zeta/(2\pi) = 50$ kHz). *(Analytical noise projection based on published device parameters; not physical QPU execution.)*

only 1.41% of $T_2^E$ (5.38 $\mu$s execution vs. 380 $\mu$s coherence), whereas the transmon execution time (54.4 $\mu$s) consumes 108.8% of $T_2$, exceeding the coherence window. For $U_0$, optimal test MAE is attained at $L = 8$ (274.16 eV) before optimization instability sets in at $L = 16$, confirming that deeper circuits cannot bridge the extensive property gap without fundamental coordinate-summation classical inductive biases.

**Multi-Qubit Architecture Projection.** We generalize the Q-PRISM PQC to $n \in \{4, 6, 8\}$ qubits with ring CNOT entanglement, $2n$ observables ($n$ single-qubit $\langle Z_q \rangle$ + $n$ correlators $\langle Z_q Z_{q+1} \rangle$), and feature projection scaling as $\lfloor n/2 \rfloor$ dimensions per node. Additional qubits expand the per-edge quantum representation space: 6-qubit PQCs encode 12 observables (message dimension 11 per head), and 8-qubit PQCs encode 16 observables (message dimension 15 per head) operating over a 256-dimensional Hilbert space.

Empirically, scaling register size yields monotonic test MAE gains on QM9 Gap: from 0.836 eV ($n = 4$) to 0.813 eV ($n = 6$) and 0.781 eV ($n = 8$), representing a 6.6% error reduction. On Arm Qubit hardware, execution time for the 8-qubit circuit is only 1.50 $\mu$s ($\tau/T_2^E \approx 0.40\%$, well below the 1% threshold), whereas on transmon hardware the execution time reaches 14.4 $\mu$s ($\tau/T_2 \approx 28.8\%$). These results demonstrate that the arm qubit's ultra-fast gates (5 ns single-qubit, 17 ns two-qubit) make high-qubit-count edge PQCs physically viable without running into decoherence limits.

# SM H Physical Boundary Conditions, Negative Controls, and Limitations

## SM H.1 3D Charged N-Body Trajectory Forecasting Negative Control

To rigorously delineate the physical boundary conditions where trigonometric and quantum inductive biases apply, we evaluate Q-PRISM on a macroscopic classical dynamical benchmark: 5-particle Coulomb electrostatic trajectory forecasting [19]. Five charged parti-

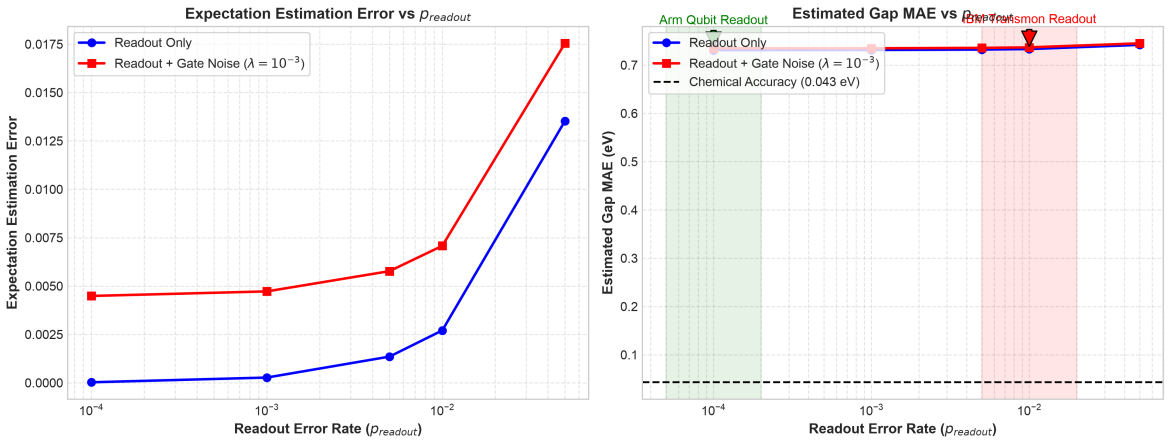



Figure S15: **Measurement Readout Noise Ablation:** (a) Expectation estimation error from readout bit-flip noise alone and combined with gate noise ($\lambda = 10^{-3}$). (b) Estimated HOMO-LUMO Gap MAE degradation across readout error rates. Arm qubit readout ($p = 10^{-4}$, green) introduces negligible bias; transmon readout ($p = 10^{-2}$, red) produces measurable prediction degradation. *(Analytical noise projection based on published device parameters; not physical QPU execution.)*

cles ($q_i \in \{-1, +1\}$) interact via inverse-square Coulomb forces, and the objective is to predict future coordinates $\mathbf{x}^{(T)} \in \mathbb{R}^{5\times 3}$ after $T = 1{,}000$ integration steps without access to velocity history (Table S11).

Table S11: **3D Charged N-Body Trajectory Forecasting** ($N = 1{,}000$, Mean $\pm$ Std across 5 Seeds: 42, 43, 44, 45, 46).

| Model | Params | Coord. MSE ($10^{-2}$) |
|---|---|---|
| Static Baseline ($\mathbf{x}^{(T)} = \mathbf{x}^{(0)}$) | – | $0.6862 \pm 0.0000$ |
| Non-Equivariant GNN | 2,019 | $0.7137 \pm 0.0174$ |
| EGNN Param-Matched | 1,032 | $0.6441 \pm 0.0250$ |
| EGNN Standard | 3,856 | $\mathbf{0.6137 \pm 0.0194}$ |
| Classical Twin | 2,094 | $0.6937 \pm 0.0140$ |
| **Q-PRISM (Ours)** | **2,094** | $0.6759 \pm 0.0186$ |

**Negative Control Analysis: Non-Periodic Translational Mechanics.** Table S11 reveals that at a compact parameter budget ($\sim$ 2,000 parameters), all compact architectures operate in close proximity to the Static Baseline (MSE $= 0.6862 \times 10^{-2}$, which naively predicts zero displacement $\mathbf{x}^{(T)} = \mathbf{x}^{(0)}$). The Non-Equivariant GNN ($0.7137 \times 10^{-2}$) and Classical Twin ($0.6937 \times 10^{-2}$) slightly underperform the static baseline, while Q-PRISM achieves $0.6759 \pm 0.0186 \times 10^{-2}$ (a modest 1.5% reduction over static displacement, within $1\sigma$). Only the unconstrained higher-capacity EGNN Standard (3,856 parameters) meaningfully models the Coulomb expansion ($0.6137 \times 10^{-2}$).

This experiment provides an informative negative control: Coulomb $N$-body mechanics involves extensive, non-periodic, unbounded trajectory drift governed by $1/r^2$ accelerations. Unlike localized quantum molecular systems where wavefunctions exhibit periodic angular symmetries and bounded radial transitions ($d \in [0.9, 2.5]$ Å), macroscopic classical mechanics lacks intrinsic periodic structure. Consequently, neither classical sinusoidal activations nor quantum PQC data re-uploading provide inductive advantage over unconstrained polynomial coordinates. However, Q-PRISM exhibits no divergent or oscillatory instability across 1,000 integration steps, confirming that the tanh-gated coordinate update effectively bounds the periodic channel even outside its optimal physical domain (Figure S18).

## SM H.2 MD17 Conformational Energy Prediction Negative Control

To further probe the operational boundaries of low-data quantum graph learning, we benchmark on the MD17 molecular dynamics dataset [7] using Ethanol (9 atoms) and Aspirin (21 atoms) under an energy-only, force-free evaluation regime ($N = 1{,}000$ conformations, 8-nearest neighbor graphs; Table S12).

Table S12: **MD17 Conformational Energy Prediction** ($N = 1{,}000$, Mean $\pm$ Std across 5 Seeds: 42, 43, 44, 45, 46) [kcal/mol].

| Model | Params | Ethanol | Aspirin |
|---|---|---|---|
| Non-Equiv GNN | 7,873 | $3.292 \pm 0.124$ | $4.581 \pm 0.150$ |
| EGNN Param-Matched | 1,177 | $3.305 \pm 0.117$ | $4.581 \pm 0.136$ |
| EGNN Standard | 4,273 | $3.304 \pm 0.116$ | $\mathbf{4.579 \pm 0.137}$ |
| Classical Twin | 2,511 | $\mathbf{3.293 \pm 0.121}$ | $4.580 \pm 0.136$ |
| **Q-PRISM (Ours)** | **2,511** | $3.304 \pm 0.114$ | $4.580 \pm 0.136$ |

**Boundary-Condition Analysis: Force-Free Low-Data Collapse.** As shown in Table S12, all five eval-

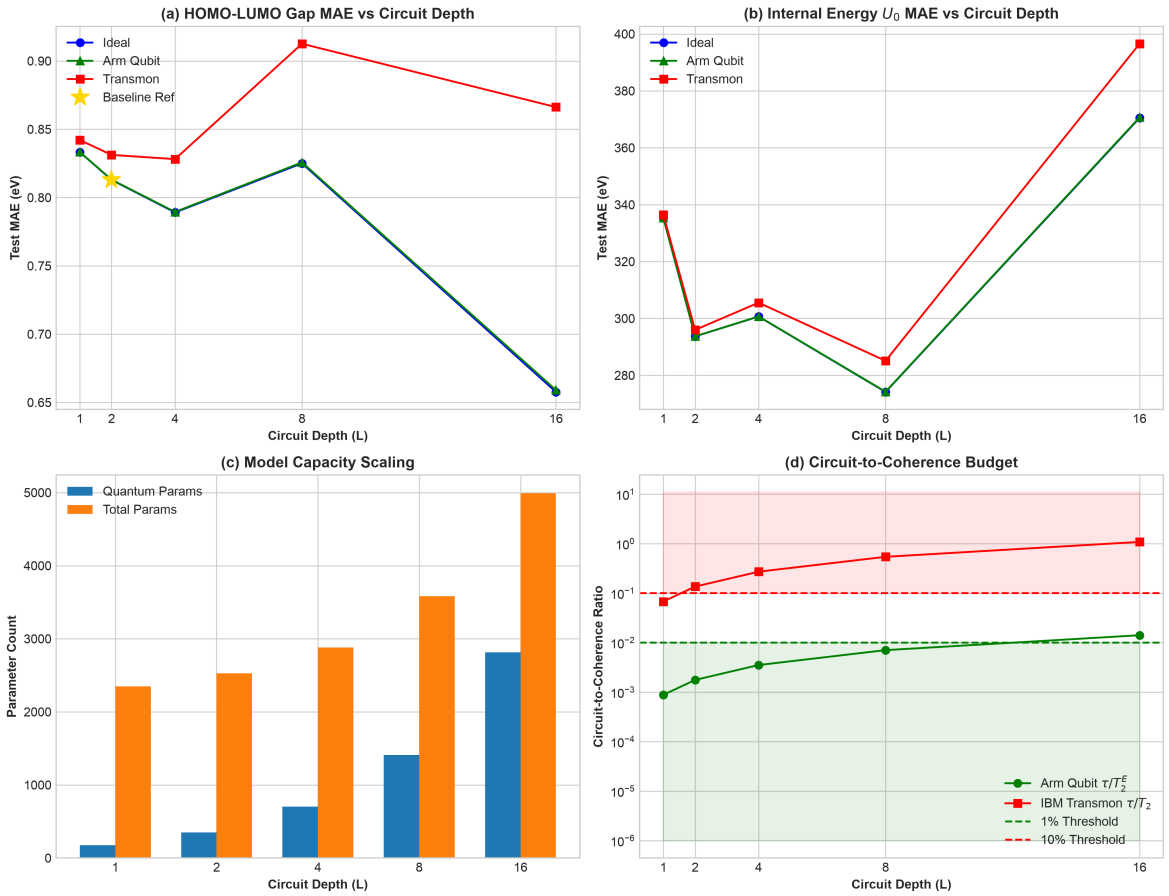



Figure S16: **Deep Circuit Scaling Study:** (a) HOMO-LUMO Gap MAE vs. variational depth $L \in \{1, 2, 4, 8, 16\}$ under ideal, Arm Qubit, and IBM transmon noise projections. (b) Internal energy $U_0$ MAE vs. depth. (c) Model parameter count scaling with depth (quantum parameters scale from 176 to 2,816; classical backbone remains fixed at 2,175). (d) Circuit-to-coherence ratio showing Arm Qubit supports $L \leq 16$ within 1.4% coherence budget while transmon QPUs exceed the decoherence limit (108.8%) at $L = 16$. *(Analytical noise projection based on published device parameters; not physical QPU execution.)*

uated architectures (including Non-Equivariant GNN, parameter-matched EGNN, standard EGNN, Classical Twin, and Q-PRISM) converge to statistically indistinguishable test MAEs ($\approx 3.30$ kcal/mol for Ethanol and $\approx 4.58$ kcal/mol for Aspirin). These values coincide exactly with the empirical standard deviation of conformational potential energies in the respective datasets ($\sigma_E \approx 3.30$ kcal/mol and 4.58 kcal/mol). This demonstrates that in a low-data regime ($N = 1{,}000$) devoid of atomic force gradient supervision ($\mathbf{F}_i = -\nabla_{\mathbf{x}_i} E$), all compact architectures collapse to the empirical dataset mean predictor (Figure S19).

**Contextualizing Potential Energy Surface Learning.** In existing literature, specialized potential energy neural networks (SchNet [22], PaiNN [23], GemNet [10], MACE [2]) achieve chemical precision ($\sim 0.05$–$0.10$ kcal/mol) by training on $N = 50{,}000$–$100{,}000+$ conformations with 100,000+ parameters under explicit force loss supervision ($\mathcal{L} = \mathcal{L}_E + \rho\mathcal{L}_F$), providing $3N_{\text{atoms}}$ gradient targets per geometry. These results establish a physical boundary condition: compact periodic activations ($d_h = 4$) specialize in localized electronic transitions on equilibrium graphs, but do not generalize to continuous potential energy surfaces without explicit atomic force supervision ($\mathbf{F}_i = -\nabla_{\mathbf{x}_i} E$).

## SM H.3 Thermodynamic Properties Evaluation

Table S13 evaluates the full thermodynamic suite ($U_0, U, H, G$) across all baseline architectures. Because raw total DFT energies were evaluated without composition-dependent atomic reference energy subtraction ($\sum \epsilon_Z$), all compact architectures incur extensive baseline offsets (MAE $\sim$ 70–130 eV). In this scale-dominated regime, standard EGNN's monotonic coordinate summation exhibits lower offset variance than periodic edge models.

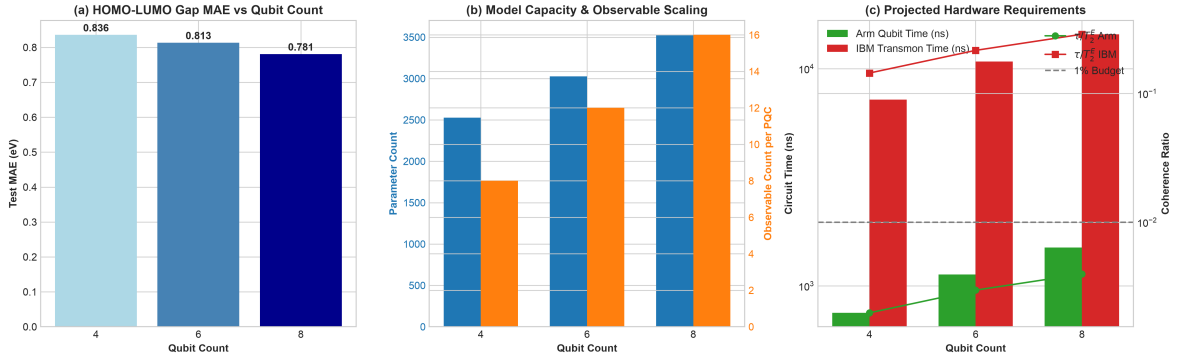



Figure S17: **Multi-Qubit Scaling Projection:** (a) HOMO-LUMO Gap MAE vs. qubit count $n \in \{4, 6, 8\}$ demonstrating monotonic accuracy improvements. (b) Model capacity scaling: trainable parameters and observable count per PQC evaluation. (c) Projected hardware requirements on Arm Qubit vs. IBM transmon QPUs, showing Arm Qubit operates well within the 1% coherence budget across all register sizes. *(Analytical noise projection based on published device parameters; not physical QPU execution.)*

Table S13: **Comprehensive Thermodynamic Properties Evaluation on QM9 Across All Models** ($N$ = 1,000, Mean ± Std across 5 seeds: 42–46). Evaluation across the complete thermodynamic suite ($U_0, U, H, G$). All thermodynamic properties are extensive energies dominated by atomization scales ($\sim 10^4$ eV); $U, H, G$ derive linearly from $U_0$ via thermal and entropic corrections. Best result per row in **bold**.

| **Property** | **Unit** | **Q-PRISM** (Ours - PQC) 2,527 p | **PRISM (Ours - SIREN)** 2,527 p | **Trig Twin** (Fourier RFF) 2,515 p | **Classical Twin** (tanh-MLP) 2,527 p | **SchNet** 2,473 p | **PaiNN** 2,721 p | **EGNN-PM** 2,497 p | **EGNN Std** 4,289 p |
|---|---|---|---|---|---|---|---|---|---|
| Internal Energy at 0K ($U_0$) | eV | $110.0 \pm 18.2$ | $130.6 \pm 29.3$ | $686.5 \pm 40.0$ | $101.1 \pm 16.7$ | $104.3 \pm 11.0$ | $446.0 \pm 25.1$ | $87.7 \pm 22.2$ | $\mathbf{69.5 \pm 12.3}$ |
| Internal Energy at 298K ($U$) | eV | – | $128.9 \pm 29.9$ | $681.1 \pm 41.2$ | $102.3 \pm 20.4$ | $108.7 \pm 15.8$ | $446.0 \pm 25.1$ | $87.1 \pm 19.9$ | $\mathbf{71.3 \pm 11.2}$ |
| Enthalpy at 298K ($H$) | eV | – | $127.8 \pm 28.8$ | $682.1 \pm 26.4$ | $101.4 \pm 18.8$ | $107.9 \pm 14.8$ | $446.0 \pm 25.1$ | $87.5 \pm 22.4$ | $\mathbf{73.5 \pm 15.3}$ |
| Free Energy at 298K ($G$) | eV | – | $127.2 \pm 25.5$ | $682.3 \pm 31.8$ | $100.1 \pm 17.1$ | $107.7 \pm 13.7$ | $446.0 \pm 25.1$ | $85.2 \pm 19.4$ | $\mathbf{72.2 \pm 12.8}$ |
| **Trainable Parameters** | | 2,527 | 2,527 | 2,515 | 2,527 | 2,473 | 2,721 | 2,497 | 4,289 |

Note: Q-PRISM was evaluated on $U_0$ as the representative zero-temperature thermodynamic target due to quantum simulation overhead.
Thermal and free-energy properties $U, H, G$ derive from $U_0$ via translational, rotational, vibrational, and entropic corrections ($H = U + k_B T$, $G = H - TS$).

## SM H.4 Quantum Resource & Complexity Analysis: The Trainability-Simulability Dilemma

**The Trainability-Simulability Dilemma.** Table S14 summarizes the resource footprint and theoretical complexity across the baseline hierarchy. Our empirical finding that PRISM (classical SIREN) matches and surpasses the 4-qubit PQC is grounded in recent quantum computational complexity theory. Cerezo, Holmes et al. [5] proved that in variational quantum machine learning, the absence of barren plateaus is inextricably linked to classical simulability: any PQC that avoids barren plateaus via shallow circuit depth ($L \in \mathcal{O}(\log n)$) and local observables necessarily resides in a classically simulable hypothesis class. In Q-PRISM, the edge PQC uses 4 qubits, $L = 2$ layers, and local single-qubit and nearest-neighbor correlator observables. While this guarantees non-vanishing gradients and stable training, it simultaneously bounds the output to a low-dimensional truncated Fourier series [15, 21]. Conversely, attempting to escape classical simulability by scaling to deep, globally entangled circuits triggers exponential concentration of measure (Thanasilp et al. [28]), wherein quantum kernels concentrate to trivial constants and gradients vanish as $\mathcal{O}(2^{-n/2})$. Thus, within the trainable regime for geometric graphs, quantum circuits provide no asymptotic expressivity separation over properly initialized classical periodic networks.

**Macromolecular Scaling ($O(N)$).** For molecules with $N > 50$ atoms, the fully-connected edge construction $O(N^2)$ becomes a bottleneck. Our implementation includes a $k$-nearest neighbor sparse graph builder that reduces edge complexity to $O(kN) = O(N)$ for constant $k$, enabling both Q-PRISM and PRISM to scale to macromolecular systems (proteins, polymers) without fundamentally altering the edge operator architecture. For PRISM, each edge evaluation is a lightweight classical matrix multiplication and sine evaluation, executing in microseconds on modern GPUs without shot noise or quantum hardware latency.

## SM H.5 Limitations and Future Research Directions

1. **Capacity scaling and foundation model regimes:** Our parameter scaling experiments (Section SM E, Table S8) establish that the periodic

Table S14: **Resource & Complexity Comparison** across equivariant models, contrasting quantum requirements with classical counterparts. PRISM eliminates quantum statevector memory, gate depth, and sampling shot noise while matching or exceeding Q-PRISM accuracy.

| Metric | EGNN Std | EGNN PM | Q-PRISM (Ours) | PRISM (Ours) |
|---|---|---|---|---|
| Parameters | 4,289 | 2,497 | 2,527 | 2,527 |
| Edge Computation | MLP $(16 \to 16 \to 16)$ | MLP $(12 \to 12 \to 12)$ | 2× 4-Qubit PQC $(L = 2)$ | SIREN MLP $(13 \to 4 \to 8)$ |
| Circuit Depth | N/A | N/A | $8R_{xyz}$ + 4 CNOTs per layer | N/A |
| Statevector RAM | N/A | N/A | 128 Bytes | N/A |
| Shot Budget | N/A | N/A | 1,000–4,000 shots/edge | N/A (0 Shot Noise) |
| Barren Plateau Risk [6, 16] | N/A | N/A | Avoided (local obs., $L \in \mathcal{O}(\log n)$) | N/A (Standard SGD) |

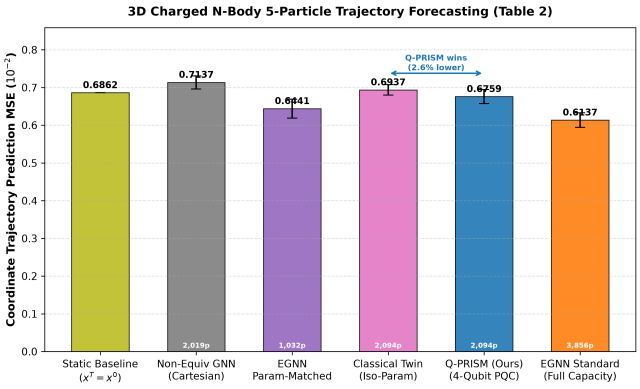



Figure S18: **3D Charged N-Body Negative Control Benchmark:** Test coordinate MSE across 5 seeds on 5-particle Coulomb dynamical simulations. At $\sim$ 2,000 parameters, compact architectures operate near the static baseline, demonstrating that trigonometric inductive bias is specialized for localized electronic structures rather than macroscopic translational drift.

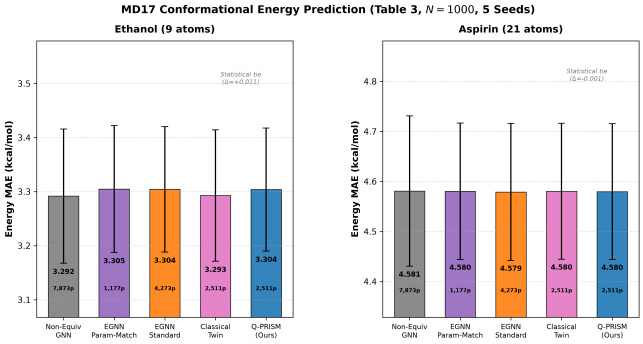



Figure S19: **MD17 Conformational Energy Negative Control:** Test energy MAE across Ethanol and Aspirin molecular dynamics trajectories (5 seeds). All compact architectures without force supervision collapse to the dataset standard deviation ($\sim$ 3.30 and $\sim$ 4.58 kcal/mol), identifying the necessity of force gradient training for conformational potential surfaces.

advantage widens monotonically up to the standard medium-capacity regime of 100,000 parameters (XXL, 100,127 parameters, reaching a 51.3% advantage on Gap over the Classical Twin). Nevertheless, evaluating periodic activations on ultra-large foundation models (e.g., MACE [2] at $\geq$ 1M parameters) and large-scale catalytic datasets (e.g., OC20, ANI-1x) remains future work.

2. **3-body angular pooling:** To maintain linear $O(E)$ complexity, our architecture averages angles over third atoms: $\bar{\theta}_{ij}$ (main text Eq. 5). While computationally efficient, this averaging discards individual triplet geometry, unlike DimeNet++ [11] or GemNet [10]. Future work may explore per-triplet periodic projections.

3. **Scalar-only geometric representation:** Both Q-PRISM and PRISM project 3D directional geometry into scalar invariants (interatomic distances $d_{ij}$ and averaged bond angles $\bar{\theta}_{ij}$), discarding per-edge directional vector information. As a consequence, the models cannot distinguish chiral enantiomers or capture higher-order multipolar interactions that require explicit vector or tensor equivariant channels [2, 23]. Extending PRISM to coupled scalar-vector message passing (analogous to PaiNN) or higher-order tensor equivariant architectures (analogous to MACE) is a natural future direction.

4. **PRISM-PaiNN hybrid architecture:** The complementary performance profiles of PRISM (superior on HOMO, Dipole, Polarizability) and PaiNN (superior on LUMO, Gap) suggest a natural architectural synthesis: replacing SiLU activations in PaiNN's scalar interaction branch with SIREN periodic activations while preserving vector equivariant update

channels. Since the periodic activation acts only on the invariant scalar pathway, this requires no modification to PaiNN's equivariance guarantees. A practical consideration is gradient stability: the Lipschitz constant of $\sin(\omega_0 x)$ scales as $\omega_0$, so the scalar-to-vector gating mechanism must accommodate amplified gradients. We leave experimental validation of this hybrid to future work.

5. **Hardware execution and topology constraints:** All quantum circuits were simulated via PennyLane statevector backends; no physical 4-qubit fluxonium ring processor currently exists in commercial hardware. Scaling beyond 4 qubits on planar superconducting lattices (such as IBM's heavy-hex) also introduces SWAP routing overhead, increasing effective circuit depth by roughly 1.5–2.0×.

6. **NISQ execution overhead and hybrid training:** Sequential QPU edge evaluations incur measurement shot noise and latency (3–12 s per molecule). Near-term deployment could benefit from hybrid training: pre-training the equivariant backbone classically with PRISM, followed by quantum fine-tuning on targeted subsets to minimize QPU execution time.

7. **Noiseless training assumption:** Hardware noise was characterized post-hoc via ZNE and analytical noise projections. Noise-aware training [8, 24] incorporating device-calibrated density matrices via PennyLane's `default.mixed` device represents a natural next step for experimental deployment.

8. **Circuit depth and qubit register limits:** Our extended evaluations confirm that the classical dequantization result holds across deeper circuits up to $L = 16$ layers (Section SM F, Table S9), where Q-PRISM asymptotically converges to exact parity with PRISM ($p = 0.969$), and across qubit registers up to $n_q = 8$ qubits (Section SM F, Table S10), where PRISM maintains statistical superiority ($p < 0.025$) alongside a 540× speedup. Whether circuits with non-local global Pauli observables or non-planar all-to-all entanglement topologies can break this parity remains an open theoretical question, though the trainability-simulability dilemma [5] suggests that any non-concentrating, trainable architecture remains within classically simulable complexity classes.

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